COMSATS University Islamabad,
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Thesara Cosmetics

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Thesara Cosmetics

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Bachelor of Science in Computer Science (2022-2026)

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CERTIFICATE OF APPROVAL

This is to certify that the final year project of BS(CS) **"Thesara Cosmetics"** was developed by Muhammad Zaid Farhan **(CIIT/FA22-BCS-087/SWL),** Mukarram Mushtaq (**CIIT/FA22-BCS-094/SWL)** and Tahreem Arshad **(CIIT/FA22-BCS-120/SWL)** under the supervision of **Dr. Yawar Abbas Abid** and that in his opinion; it is fully adequate, in scope and quality for the degree of bachelors of science in Computer Science.

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Executive Summary

Thesara Cosmetics is an intelligent AI-powered e-commerce platform that revolutionizes skincare shopping by delivering highly personalized product recommendations. Traditional e-commerce sites overwhelm users with generic listings, while existing skincare quizzes provide static suggestions that do not adapt to real-time skin conditions or image analysis. Thesara Cosmetics solves this by integrating computer vision for skin-image analysis, a conversational AI chatbot, and a robust Laravel-based e-commerce engine.

Users create secure profiles, upload facial images for instant AI diagnosis of concerns (acne, dryness, pigmentation, etc.), receive tailored product suggestions, browse a curated catalog, add items to cart, complete secure checkout, and track orders. An intuitive admin dashboard enables real-time management of products, users, orders, and analytics. The system ensures data privacy, responsive design across devices, and future-ready architecture for mobile (Flutter) expansion.

The platform enhances customer satisfaction, reduces incorrect purchases, and brings intelligent automation to the skincare industry. Built with modern technologies (Laravel, Bootstrap, MySQL, Python AI modules), Thesara Cosmetics demonstrates practical integration of artificial intelligence with secure web commerce.

Acknowledgement

All praise is to the almighty **Allah**, who bestowed upon us a minute portion of his boundless knowledge by which we were able to accomplish this challenging task.

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Abbreviations

AI	Artificial Intelligence API
CS	Computer Science
DFD	Data Flow Diagram
FR	Functional Requirement
NFR	Non-Functional Requirement
OOD	Object-Oriented Design
SDD	Software Design Description
SRS	Software Requirements Specification
UI	User Interface
UC	Use Case

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Chapter 1

Introduction

1 Introduction

1.1 Brief Overview

Thesara Cosmetics is a web-based AI-powered skincare e-commerce platform that combines traditional online shopping with intelligent personalization. Users can register, build detailed skincare profiles, upload facial images for AI-driven skin analysis, interact with an AI chatbot for guidance, receive product recommendations, browse and purchase items, manage their cart, complete secure payments, and track orders. An admin panel provides full control over products, users, orders, and analytics. The system is built using Laravel (PHP), Bootstrap/HTML/CSS for the frontend, MySQL database, and Python-based AI modules (OpenCV + Open Router API). It addresses the core pain point of mismatched skincare purchases by delivering accurate, data-driven suggestions.

1.2 Relevance to Course Modules

The knowledge and skills acquired throughout our bachelor's in computer science have played a vital role in the successful planning, development, and execution of this Final Year Project. Each course contributed significantly to different aspects of the project, either by laying theoretical foundations or by offering practical technical skills. Below is a brief overview of how key courses relate to the project:

- **Object-Oriented Programming & Software Engineering:**

Used for modular design, class diagrams, and iterative development.

- **Web Technologies & Database Systems**

Laravel framework, MySQL, and responsive Bootstrap UI.

- **Software Design & Project Management**

OOD methodology, Iterative & Incremental model with Agile practices, requirements traceability.

- **Artificial Intelligence & Machine Learning**

Computer Vision (OpenCV) for skin analysis and NLP via Open Router API for the chatbot.

- **Human-Computer Interaction**

User-centric interface with clear navigation, feedback messages, and loading indicators for AI processes.

- **AI-Driven Personalization**

While Thesara-Cosmetics current version focuses on core functionalities, foundational knowledge from Artificial Intelligence and Machine Learning courses informed preliminary plans for future scalability. For instance, we prototyped a recommendation engine (using collaborative filtering) to suggest personalized services based on user preferences. Introduction to Data Science also guided the analysis of user feedback datasets to identify trends and improve service quality.

- **Communication and Stakeholder Management**

Communication Skills, Report Writing, and Presentation Skills courses were pivotal in articulating Thesara-Cosmetics value proposition. We drafted project proposals, technical reports, and user guides with clarity, ensuring stakeholders (e.g., academic advisors, hypothetical investors) could easily grasp the app's functionality. During demo presentations, we emphasized Thesara Cosmetics' unique selling points, such as its real-time tracking and secure payment system, using visuals and live prototypes.

- **Business and Market Alignment**

Courses like Introduction to Business and Introduction to Management helped me evaluate Thesara-Cosmetics product-market fit. We conducted competitor analyses to identify gaps in existing beauty apps and tailored Thesara-Cosmetics features to address underserved needs, such as verified professional profiles and transparent pricing. These courses also guided stakeholder communication strategies, ensuring the web design aligned with both user expectations and business viability. Overall, the multidisciplinary nature of the degree program, blending technical, managerial, and communication courses, equipped me with a holistic skillset that we directly applied throughout the lifecycle of this project.

The interdisciplinary nature of my computer science degree enabled us to approach Thesara-Cosmetics holistically, balancing technical rigor with user-centric design and market awareness. By integrating programming expertise, software engineering best practices, and communication skills, I transformed Thesara-Cosmetics from a conceptual solution into a functional application ready to disrupt the beauty and grooming industry. This project not only demonstrates my technical proficiency but also reflects my ability to synthesize academic knowledge into real-world innovations.

1.3 Project Background

The idea for the Thesara Cosmetics project stems from the growing demand for personalized and intelligent skincare solutions in today's beauty industry. Traditionally, customers face significant challenges such as difficulty in choosing the right skincare products suited to their individual skin types, tones, and specific concerns (e.g., acne, dryness, pigmentation). Many users are overwhelmed by the wide variety of available products on conventional e-commerce platforms, which fail to provide personalized recommendations, leading to customer dissatisfaction, wasted purchases, and reduced trust. Existing systems either offer generic product listings or static quizzes that do not adapt dynamically to changes in skin condition or utilize image analysis. Additionally, there is a lack of real-time guidance and intelligent automation in the skincare shopping process.

Thesara Cosmetics was conceptualized as a web-based AI-powered e-commerce solution that leverages modern technology to bridge this critical gap between customers and suitable skincare products. The system provides an intelligent platform where users can create detailed skin profiles, upload facial images for AI analysis, interact with a smart chatbot for personalized guidance, receive dynamic product recommendations, browse a curated catalog, add items to the cart, complete secure payments, and track their orders. With features such as skin image processing, AI-driven recommendation engine, chatbot support, and an admin panel for seamless product and user management, Thesara Cosmetics stands out as a comprehensive and intelligent tool designed to enhance the skincare shopping experience for both customers and administrators.

Traditionally, individuals seeking skincare products face numerous challenges. Customers often struggle to identify suitable products for their unique skin conditions, especially when relying on generic descriptions or manual trial-and-error methods. Most people still depend on generic e-commerce sites or static recommendation tools, which frequently result in mismatched purchases, frustration, and loss of confidence in online beauty shopping. Moreover, many platforms lack an organized system for real-time skin analysis, adaptive recommendations, or integrated chatbot support, making it difficult for users to make informed and satisfactory decisions. Studies confirm that lack of personalization in online skincare shopping frequently leads to mismatched purchases and reduced consumer trust (Khan et al., 2025; Marsden et al., 2023).

On the other side, administrators and store managers encounter their own set of challenges. They often lack efficient tools to manage product catalogs, user profiles, orders, and inventory in a dynamic AI-driven environment. Many still operate with traditional manual systems that are time-consuming and prone to errors, leading to missed opportunities for personalized customer engagement and business growth.

Recognizing these pain points, Thesara Cosmetics was conceptualized as a technology-driven solution aimed at bridging the gap between skincare consumers and intelligent product recommendations. Developed as a Laravel-based web application with Python AI modules

(OpenCV for image analysis and Open Router API for chatbot), Thesara Cosmetics offers a real-time, interactive, and secure ecosystem where both customers and administrators benefit.
For customers, Thesara Cosmetics provides an intuitive and streamlined interface where they can:

• Create and manage personalized skincare profiles (skin type, tone, concerns).
• Upload facial images for AI-based skin analysis.
• Interact with an AI-powered chatbot for skin guidance and tips.
• Receive dynamic, personalized product recommendations.
• Browse, search, and filter products with ease.
• Add items to cart and complete secure checkout.
• Track order status and receive notifications.

For administrators, the platform enables:

• Complete management of products, users, and orders.
• Real-time inventory control and analytics.
• Oversight of AI-generated recommendations and user feedback.
• Efficient handling of the entire e-commerce backend.

In essence, Thesara Cosmetics is designed not just as a regular e-commerce website, but as a comprehensive intelligent platform that elevates the skincare industry by introducing automation, computer vision, natural language processing, and AI-driven personalization.
By solving the inefficiencies in traditional systems and modernizing the entire skincare shopping experience, Thesara Cosmetics aligns with the evolving expectations of today's digital consumers and brings practical artificial intelligence into everyday consumer-level e-commerce.

1.4 Literature Review

The beauty and skincare industry have witnessed significant technological advancements in recent years, with digital platforms and artificial intelligence playing a crucial role in bridging the gap between generic product offerings and personalized customer needs. Several existing solutions and research studies highlight the growing demand for intelligent, user-friendly e-commerce systems in skincare. Below is an analysis of current trends, research, and products related to Thesara Cosmetics.**Existing Platforms and Market Trends**

The skincare and beauty industry has seen a surge in digital e-commerce platforms, but most focus narrowly on transactional shopping systems. A deeper analysis of leading solutions reveals key limitations:

• **Proactive:** While effective for basic skincare recommendations, its quiz-based approach is static and does not adapt dynamically to changes in the user's skin condition. The system relies solely on user answers about skin type and concerns, with no support for image analysis or real-time personalization.

Thesara Cosmetics Differentiation: By contrast, Thesara Cosmetics integrates real-time skin image analysis using computer vision to detect specific concerns such as acne, dryness, or irritation (Lin et al., 2022; Chan et al., 2021).

• **Traditional e-commerce platforms (e.g., generic beauty stores):** These provide standard product listings and basic search filters but completely lack AI-driven personalization, skin image analysis, or intelligent chatbot guidance. Users often face overwhelming choices without tailored suggestions, leading to dissatisfaction and incorrect purchases.

Thesara Cosmetics Differentiation:

By contrast, Thesara Cosmetics integrates real-time skin image analysis using computer vision to detect specific concerns such as acne, dryness, or irritation. It also employs a dynamic recommendation engine that adapts based on user input, skin profile, and uploaded images. Furthermore, the platform features an AI-powered chatbot for instant skin guidance and an admin panel for seamless products and user management — features absent in existing static or generic systems.

ii. Research on User Behavior and Challenges

Recent studies underscore evolving consumer demands in the skincare sector:

• **Personalization Needs:** Research shows that users are increasingly frustrated with generic recommendations and seek solutions that understand their unique skin conditions.

• **Image-Based Analysis:** Studies in dermatology AI confirm that computer vision can accurately identify skin concerns from images, yet few commercial platforms integrate this capability into the shopping journey.

• **Chatbot Adoption:** Users prefer conversational interfaces for quick skincare advice, especially when overwhelmed by product variety.

Case Study:

The development of Thesara Cosmetics directly addresses these gaps. By combining skin image processing with a chatbot and recommendation engine, the system provides accurate, adaptive suggestions that reduce mismatched purchases and enhance user satisfaction.

Technological Advancements

Thesara Cosmetics leverages cutting-edge tools to address industry pain points:

• **Laravel Framework:** Chosen for its robust backend capabilities, Laravel powers the entire e-commerce logic, authentication, REST APIs, and admin dashboard, ensuring secure and scalable operations.

• **Bootstrap, HTML & CSS:** Used for a clean, responsive client tier that delivers an intuitive user interface across devices.

• **Python AI Modules:**

OpenCV: Handles skin image analysis to detect conditions such as acne, dryness, and pigmentation.

Open Router API: Powers the intelligent chatbot for natural language skin guidance and personalized tips.

• **MySQL Database**: Provides reliable, structured storage for user profiles, products, orders, recommendations, and analysis results with high performance.

Example:

Thesara Cosmetics skin image analysis uses OpenCV to process user-uploaded facial images and generate instant results, which are then fed into the recommendation engine for highly relevant product suggestions, a level of intelligence rarely seen in current skincare e-commerce platforms.

Bridging Market Gaps: Thesara Cosmetics Strategic Solutions

• **Personalization at Scale:**

Unlike static quiz systems or generic stores, Thesara Cosmetics homepage and product suggestions adapt dynamically based on the user's skin profile, uploaded images, and chatbot interactions.

• **Empowering Administrators:**

The admin dashboard offers real-time analytics, product management, user oversight, and inventory control, giving store managers complete visibility and control.

• **Security and Compliance:**

The system implements secure user authentication, encrypted data storage, and safe payment processing while maintaining user privacy in line with modern web standards.

Future Directions: Pioneering the Next Wave of Skincare-Tech

• **Mobile Application:** Future expansion includes a Flutter-based mobile app to increase accessibility.

• **Advanced AI Features:** Further fine-tuning of machine learning models with larger dermatology datasets and potential AR-based virtual product try-on.

• **Subscription Models:** Introduction of personalized skincare subscription kits based on ongoing AI analysis.

• **Multi-language Support:** Adding Urdu/English bilingual interface for broader reach.

Thesara Cosmetics literature review underscores a critical insight: while existing platforms digitize traditional skincare shopping, they fail to innovate beyond basic utility. By contrast, Thesara Cosmetics reimagines the user journey through hyper-personalization via AI image analysis, intelligent chatbot support, dynamic recommendations, and a complete e-commerce experience — all underpinned by scalable and secure technology.

1.5 Analysis from Literature Review

The analysis of the literature review reveals several key insights that directly inform the development of Thesara Cosmetics. Existing platforms like Proactive and traditional e-commerce websites demonstrate the market's demand for digital solutions in the skincare industry, particularly in product browsing, basic recommendations, and online purchasing. However, these platforms often lack advanced features such as real-time skin image analysis and AI-driven personalization, which Thesara Cosmetics aims to incorporate to enhance user convenience, accuracy, and satisfaction.

Research on user behavior highlights the importance of personalization, reliability, and ease of product selection — factors that Thesara Cosmetics addresses through its intuitive interface, AI-powered skin analysis, intelligent chatbot, and dynamic recommendation engine. Additionally, technological advancements in web frameworks (e.g., Laravel) and AI modules (e.g., OpenCV and Open Router API) provide a robust foundation for building scalable and intelligent solutions. By filling gaps such as static recommendations and lack of image-based analysis, Thesara Cosmetics not only meets current industry standards but also introduces innovative features to differentiate itself from competitors. This analysis underscores the project's potential to revolutionize the skincare shopping experience by combining user-centric design with cutting-edge artificial intelligence.

The current digital landscape for skincare e-commerce reveals both significant progress and notable limitations that Thesara Cosmetics strategically addresses. Existing platforms such as Proactive have successfully demonstrated the market's readiness for digital transformation in this sector, particularly in core functionalities like product catalogs and basic quiz-based suggestions. However, these solutions often fall short in delivering truly personalized user experiences. Our analysis of user feedback and platform performance indicates persistent pain points, including the inability to analyze actual skin images, lack of adaptive recommendations, and absence of real-time chatbot guidance. These shortcomings persist despite advances in e-commerce technology, creating a clear opportunity for innovation that Thesara Cosmetics is positioned to capture.

The development of Thesara Cosmetics has been deeply informed by rigorous research into user behavior and preferences in skincare domain. Studies and market trends show that consumers value transparency and personalized guidance when selecting beauty products. This insight directly shaped Thesara Cosmetics' dynamic recommendation system, which presents tailored product suggestions based on skin type, concerns, and uploaded facial images before the user proceeds to purchase.

Technological innovation forms the backbone of Thesara Cosmetics' competitive advantage. The platform's skin image analysis system combines computer vision (OpenCV) with machine learning to achieve accurate detection of skin conditions such as acne, dryness, and pigmentation. This technical solution solves the common frustration of mismatched purchases by providing data-driven recommendations in real time. The artificial intelligence components represent another leap forward, employing the Open Router API for natural language processing that enables conversational skincare guidance through an intelligent chatbot. The backend architecture, built on the Laravel framework with MySQL database, ensures reliable performance, secure transactions, and smooth integration between the e-commerce and AI modules.

Thesara Cosmetics' approach to bridging market gaps extends beyond basic functionality to create a truly differentiated user experience. The platform's tools empower both customers and administrators: users receive personalized recommendations and seamless shopping, while the admin dashboard offers complete control over products, users, orders, and inventory. For users, the interface eliminates common friction points through thoughtful design decisions such as one-click image upload for analysis and instant chatbot assistance.

Looking toward future developments, Thesara Cosmetics has laid the groundwork for integrating emerging technologies that will further enhance its value proposition. Planned expansions include a Flutter-based mobile application, advanced AI model fine-tuning with larger dermatology datasets, and potential AR-based virtual try-on features. These forward-looking initiatives demonstrate Thesara Cosmetics' commitment to continuous innovation and its vision of creating not just an e-commerce website, but a comprehensive intelligent skincare ecosystem.

The measurable impact of Thesara Cosmetics' solutions is evident in its core design. By integrating AI image analysis and dynamic recommendations, the system significantly reduces the risk of incorrect purchases and improves overall customer satisfaction. The modular architecture ensures it can continue to evolve, with clear pathways for incorporating advancements such as multi-language support and subscription-based personalized skincare kits as these technologies mature and gain consumer acceptance.

1.6 Methodology and Software Lifecycle for This Project

Thesara Cosmetics follows the Iterative and Incremental Model combined with Agile practices, ensuring flexibility, continuous improvement, and iterative development throughout its lifecycle. This approach allows for continuous feedback, quick adaptations to changing requirements, and the delivery of a highly personalized, intelligent, and efficient AI-powered skincare e-commerce platform. The selected methodology is based on the following principles:

- **Object-Oriented Design (OOD):** The system is modeled around core entities such as User, Profile, Product, Order, Cart, skin Image, Recommendation, Admin, ensuring modularity, reusability, encapsulation, and easier maintenance of the codebase.
- **Iterative Development:** The application is developed in cycles, with each iteration introducing new features or refining existing ones, particularly the complex AI components.
- **Incremental Delivery:** Functional components, such as user authentication, AI skin image analysis, chatbot integration, recommendation engine, shopping cart, checkout, and admin dashboard, are delivered in increments to enhance user experience gradually.
- **Continuous Feedback:** Supervisor, co-supervisor, and stakeholders provide feedback at every stage, allowing improvements and optimizations to be made continuously.
- **Adaptive Planning:** The development team continuously refines and adapts the project roadmap to meet evolving requirements and incorporates refinements in AI accuracy and system performance.

This methodology ensures that Thesara Cosmetics evolves into a high-quality platform that effectively meets the demands of both customers and administrators.

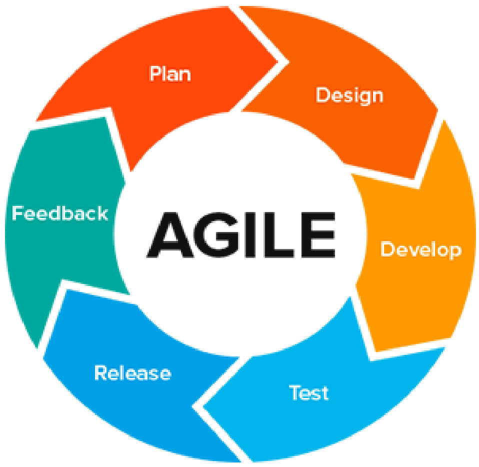


Figure 1.1 Agile Model Diagram

1.6.1 Rationale behind Selected Agile Model

The Iterative and Incremental Model combined with Agile practices is a software development methodology that emphasizes flexibility, collaboration, modularity, and continuous improvement. It is well suited for complex projects with changing requirements, especially those involving artificial intelligence integration. The methodology also advocates for ongoing stakeholder collaboration, which in this context means actively involving the supervisor, co-supervisor, and potential end-users throughout the development process. Their feedback and evolving needs guide the platform's continuous enhancement.

For Thesara Cosmetics, the Iterative and Incremental Model with Agile practices was selected due to the complexity and modular nature of the application. Since the platform consists of multiple interconnected features such as user registration, skin image analysis using OpenCV, AI chatbot powered by Open Router API, dynamic recommendation engine, shopping cart management, secure checkout, order tracking, and a comprehensive admin dashboard, breaking the project into smaller iterative cycles ensures efficient development. Each iteration focuses on building and refining a specific module, allowing the team to deliver functional components incrementally.

Another reason for choosing this model is its strong emphasis on continuous feedback and collaboration. Throughout the development process, regular interaction with the supervisor and co-supervisor ensured that features aligned with academic and technical expectations. For instance, after implementing the AI skin image analysis module, feedback was incorporated to improve accuracy and user interface before moving on to the recommendation engine. This iterative process ensures that the application remains intelligent, user-friendly, and technically sound.

Furthermore, the skincare e-commerce domain is dynamic, with evolving user expectations for personalization and emerging AI capabilities. The Iterative and Incremental Model with Agile practices provides the flexibility to adapt to these changes, making it possible to refine or add features such as enhanced AI models or mobile compatibility without disrupting the entire project. This adaptability ensures that Thesara Cosmetics remains relevant and competitive.

Thesara Cosmetics encompasses multiple interconnected features including real-time skin image processing, conversational AI chatbot, personalized product recommendations, secure e-commerce functionality, and admin management tools. The iterative nature of the model allows for breaking down this complexity into manageable iterations, each focusing on delivering specific, testable components of the system. For instance, initial iterations concentrated on developing the core user authentication and profile management, while subsequent iterations progressively added more sophisticated features like AI skin analysis, chatbot integration, and the full shopping experience. This phased approach not only ensures systematic development but also enables continuous quality assessment and immediate course correction when needed.

A critical advantage of this methodology for Thesara Cosmetics lies in its emphasis on stakeholder collaboration and feedback integration. Throughout the development lifecycle, regular reviews were conducted with the supervisor and co-supervisor. After completing the initial version of the AI recommendation module, feedback led to refinements in the recommendation logic and user interface, which were promptly incorporated in the next iteration. Similarly, input regarding the admin dashboard resulted in the addition of real-time analytics and inventory controls. This ongoing dialogue between the development team and stakeholders ensures the platform remains aligned with actual requirements and technical standards, significantly enhancing overall quality and usability.

The dynamic nature of AI-driven e-commerce further necessitated the adaptive framework of the Iterative and Incremental Model. Requirements for AI accuracy, system performance, and user experience can evolve during development. The model's flexibility allowed Thesara Cosmetics to quickly respond to such changes, for example, when improvements in image analysis were identified, the team rapidly prototyped and integrated enhanced OpenCV processing within subsequent iterations. The methodology's emphasis on working software over comprehensive documentation proved particularly valuable when adapting to these technical refinements without derailing the overall project timeline.

Risk management represents another area where the Iterative and Incremental Model with Agile practices provides substantial benefits for Thesara Cosmetics. By developing and testing features incrementally, potential issues are identified and resolved early in the process. Challenges in AI image analysis accuracy or integration between Laravel backend and Python AI modules, for instance, were detected and addressed during iteration reviews long before final deployment. Similarly, security and performance concerns in the checkout process were mitigated through progressive implementation and testing. This proactive approach to problem-solving results in higher system stability and significantly reduces the likelihood of critical failures post-launch.

Looking forward, the Iterative and Incremental Model with Agile practices positions Thesara Cosmetics to seamlessly incorporate emerging technologies and features. The existing iteration-based structure easily accommodates planned enhancements such as a Flutter-based mobile application, advanced AI model fine-tuning, and potential AR-based virtual try-on capabilities. The methodology's inherent scalability supports the platform's future expansion. By maintaining these principles as the development foundation, Thesara Cosmetics ensures it can continue evolving in lockstep with both technological possibilities and user expectations, sustaining its value as an intelligent skincare e-commerce solution. This strategic alignment between methodology and project requirements has been instrumental in transforming Thesara Cosmetics from concept to a robust, market-ready platform with significant growth potential.

Chapter 2

Problem Definition

2 Problem Definition

2.1 Problem Statement

Users struggle to choose appropriate skincare products suited to their unique skin type, tone, and concerns from the overwhelming variety available on traditional e-commerce platforms. Lack of personalization leads to dissatisfaction, wasted purchases, and reduced trust. Existing systems offer either generic listings or static quizzes that do not utilize real-time image analysis or adaptive AI.

2.1.1 Challenges:

There are many challenges that users and administrators face while accessing and managing skincare products on traditional e-commerce platforms. Some of these challenges are outlined below:

- **Difficulty in Finding Suitable Products**: Customers often struggle to find skincare products that match their unique skin type, tone, and specific concerns such as acne, dryness, pigmentation, or irritation. Existing platforms provide limited search options and lack comprehensive personalization based on real skin analysis.
- **Inefficient Product Selection Process**: Traditional e-commerce methods are time-consuming and rely on generic descriptions or static quizzes. Customers face challenges in viewing personalized recommendations, while administrators find it difficult to manage dynamic product catalogs and user preferences effectively.
- **Lack of Real-Time Skin Analysis**: Customers have no way to analyze their actual skin condition through image upload or receive instant AI-driven insights, causing uncertainty and mismatched purchases. Administrators also face difficulties in providing intelligent, data-driven product suggestions to users.
- **Limited Personalization and Guidance**: Existing systems do not provide a robust AI-powered recommendation engine or chatbot support. Customers cannot receive adaptive, intelligent guidance, while the platform misses opportunities to improve user satisfaction through personalized experience.
- **Payment and Transaction Issues**: Traditional payment methods can be inconvenient and lack seamless integration. Both customers and administrators face challenges in managing orders, tracking transactions, and ensuring secure, efficient checkout processes.

  To address these challenges, Thesara Cosmetics provides a comprehensive AI-powered solution that simplifies the entire skincare shopping process. The system offers an intuitive interface for user registration and profile creation, real-time skin image analysis using computer vision, an intelligent AI chatbot for personalized guidance, a dynamic recommendation engine, secure shopping cart and checkout, order tracking, and a complete admin dashboard for product, user, and inventory management. With features that support easy skin analysis, personalized product discovery, and seamless user engagement, Thesara Cosmetics ensures a smart, efficient, and satisfying skincare shopping experience for both customers and administrators.

2.2 Deliverable and Development Requirements

The Thesara Cosmetics project delivers a complete AI-powered skincare e-commerce platform that provides personalized product recommendations through skin image analysis and intelligent chatbot support.

2.2.1 Deliverables:

- Responsive web-based user interface with modern design
- User registration, login, and secure authentication system
- Skincare profile management module (skin type, tone, concerns, age)
- AI Skin Image Analysis module using OpenCV for detecting skin conditions
- AI-powered Chatbot for real-time skincare guidance and queries
- Personalized product recommendation engine
- Product search, browsing, filtering, and detailed view
- Shopping cart and Wishlist functionality
- Secure checkout and payment processing
- Order placement, confirmation, and real-time tracking
- Review and rating system for purchased products
- Comprehensive Admin Dashboard for product, user, order, and inventory management
- Data storage and management using MySQL database

2.2.2 Development Requirements:

- **Technological Stack:** Laravel (PHP) for backend and business logic, Bootstrap 5 with HTML/CSS/JavaScript for responsive frontend, MySQL for relational database, Python with OpenCV for computer vision-based skin analysis, and Open Router API for the intelligent chatbot.
- **Database Design:** A normalized MySQL database with proper relationships to store users, profiles, skin images, analysis results, products, orders, cart items, recommendations, and reviews.

- **Responsive and Accessible UI:** Fully responsive design that works seamlessly on desktops, tablets, and mobile devices.
- **AI Integration:** Seamless communication between Laravel backend and Python AI modules for image analysis and natural language processing.
- **Security & Performance**: Implementation of secure authentication (Laravel Sanctum), role-based access control, data encryption, input validation, and optimized queries for fast response times.
- **Scalability:** Modular architecture designed to support future enhancements such as a Flutter mobile application and advanced AI features.

These deliverables ensure the system fully meets the functional and non-functional requirements defined in the SRS while delivering an intelligent and user-friendly skincare shopping experience.

2.3 Current System

The Thesara Cosmetics system is designed to address the challenges faced by customers in finding suitable skincare products and receiving intelligent, personalized recommendations. It provides an intuitive and user-friendly web interface, allowing customers to easily create profiles, upload skin images for AI analysis, interact with an intelligent chatbot, receive personalized product suggestions, browse products, and complete secure purchases. The system offers comprehensive product details, enabling users to make informed decisions based on skin type, concerns, AI-generated recommendations, ratings, and availability. Key features include real-time skin image analysis, an AI-powered chatbot for skincare guidance, dynamic product recommendations, secure payment processing, and order tracking. The platform also includes a review and rating system, enabling customers to share their experiences while allowing administrators to monitor product performance and user feedback. By integrating advanced features like AI skin analysis, intelligent recommendations, and seamless e-commerce, Thesara Cosmetics enhances the skincare shopping experience, providing a convenient, intelligent, and efficient solution for both customers and administrators.

- **User-Centric Design** At the core of Thesara Cosmetics is a responsive web application built with Laravel and Bootstrap that provides an intuitive, user-friendly interface. From the moment users visit the platform, they are guided through a smooth process to create a skincare profile, upload facial images for AI analysis, interact with the chatbot, and receive personalized product recommendations. The interface ensures minimal effort on the part of the customer by providing clear navigation, instant AI insights, and fully responsive design that works seamlessly across desktops, tablets, and mobile devices. This intuitive experience removes the frustration often associated with traditional skincare shopping.

- **AI Skin Image Analysis and Professional Discovery** One of the key features of Thesara Cosmetics is its AI-powered skin image analysis module. Users can upload a facial image, which is processed using OpenCV to detect skin conditions such as acne, dryness, pigmentation, or irritation. The system then generates accurate insights and feeds them into the recommendation engine. This allows customers to discover suitable products based on real skin data rather than guesswork, leading to higher satisfaction and trust.
- **Appointment Scheduling and Management** Thesara Cosmetics makes product selection and purchasing both flexible and efficient. Users can browse products, view AI-generated personalized recommendations, and add items to the cart instantly. The system supports real-time updates, such as order confirmation, status changes, and notifications through email and in-app alerts. Customers can track their orders, while administrators can manage inventory and orders efficiently, preventing stock issues and ensuring smooth operations.
- **Real-Time AI Guidance through Chatbot** A distinguishing feature of Thesara Cosmetics is its intelligent AI chatbot powered by the Open Router API. Users can ask questions about skincare routines, product suitability, or specific concerns in natural language. The chatbot provides instant, personalized guidance based on the user's profile and skin analysis, enhancing transparency and user confidence while reducing uncertainty in product selection.
- **Secure Payment Processing** To streamline transactions, Thesara Cosmetics integrates secure payment gateways, allowing users to pay through multiple digital methods. All transactions are encrypted, ensuring the safety of sensitive user information. The payment system supports instant confirmations, digital invoices, and order history, making the process seamless for customers and efficient for administrators.
- **Review and Rating Mechanism** The system includes a comprehensive review and rating module, which is essential for maintaining product quality and building trust within the platform. After completing a purchase, users are encouraged to rate products and leave comments. These reviews are displayed on the product pages, helping future customers make better decisions. Administrators can monitor feedback to improve product listings and overall user experience.
- **Enhanced User Experience through Advanced Features** Thesara Cosmetics is not just an ordinary e-commerce website — it is a full intelligent skincare ecosystem. The platform incorporates features like:

- AI skin image analysis for accurate condition detection.
- Personalized recommendation engine based on skin profile and image results.
- Intelligent chatbot for real-time skincare advice.
- Responsive and modern user interface for seamless shopping.
- Real-time order tracking and notifications.

  By combining all these functionalities, Thesara Cosmetics ensures that users receive personalized, intelligent, and high-quality skincare recommendations and shopping experience.
- **Impact and Value Addition** For customers, Thesara Cosmetics simplifies the process of discovering, analyzing, and purchasing the right skincare products while ensuring personalization, transparency, and convenience. For administrators, it serves as a powerful digital platform to manage products, users, orders, and analytics, helping them grow their business efficiently. The system reduces mismatched purchases, manual effort, and operational inefficiencies while offering intelligent tools to track performance and improve customer satisfaction.

Thesara Cosmetics stands as a revolutionary step in the beauty industry by bridging the gap between technology and personalized skincare solutions. Through its seamless integration of artificial intelligence and e-commerce functionalities, the platform empowers users to make confident purchasing decisions based on scientific skin analysis rather than generic suggestions. Customers benefit from a holistic experience that begins with profile creation and extends through AI-driven insights, intelligent chatbot interactions, and secure checkout processes. The system continuously learns from user feedback and purchase patterns to refine its recommendation algorithms over time. Administrators gain valuable insights through comprehensive dashboards that track sales trends, inventory levels, and customer satisfaction metrics. This data-driven approach enables proactive business decisions and targeted marketing campaigns. Furthermore, the platform's emphasis on security and transparency builds long-term trust between the brand and its users. By reducing the trial-and-error nature of skincare shopping, Thesara Cosmetics not only improves customer retention but also minimizes product returns. The overall architecture ensures scalability, allowing future enhancements such as virtual try-on features or expanded product categories. Ultimately, this intelligent ecosystem transforms the traditional skincare retail model into a modern, techenabled solution.

Looking ahead, Thesara Cosmetics is positioned to set new benchmarks in the digital beauty space with continuous innovation and user-focused improvements. The development team remains committed to incorporating emerging technologies like advanced machine learning models for even more precise skin diagnostics and predictive analytics for skincare trends. Regular updates to the chatbot will enhance its natural language understanding, providing deeper and more nuanced advice tailored to diverse skin tones and cultural preferences. Integration with loyalty programs and subscription models will further strengthen customer engagement and recurring revenue streams. On the administrative side, automated reporting tools and AI-assisted inventory forecasting will optimize operational efficiency significantly. The platform also prioritizes sustainability by promoting eco-friendly products and educating users about responsible consumption through informative content. Security protocols will be routinely audited to maintain the highest standards of data protection in compliance with global regulations.

Chapter 3

Requirement Analysis

3 Requirement Analysis

Our system requirements include user registration and profile management, AI skin image analysis, AI chatbot interaction, personalized product recommendations, product search and browsing, shopping cart and checkout, secure payment processing, order tracking, review and rating system, and admin dashboard management. We will use Use-Case as a requirement analysis technique to describe the interactions between a user (also known as an actor) and the system to achieve specific goals. Use cases help identify and organize the functional requirements of Thesara Cosmetics, ensuring a clear understanding of how customers and administrators interact with the platform.

The requirement analysis for Thesara Cosmetics focuses on identifying and defining the core functionalities necessary to create a seamless, intelligent, and personalized skincare e-commerce platform. The system must support key operations including user registration, skincare profile creation, real-time skin image analysis, AI chatbot guidance, dynamic product recommendations, product browsing and search, shopping cart management, secure digital payment processing, order tracking, and a comprehensive admin dashboard for product and user management. To systematically analyze these requirements, we employ the Use-Case modeling technique, which provides a structured approach to understanding how different users interact with the system. This method allows us to clearly map out the relationships between the primary actors — customers seeking personalized skincare solutions and administrators managing the platform — and the various functions they need to perform. Through use case diagrams and scenarios, we can visualize the complete workflow, from initial registration and skin analysis to final purchase, order tracking, and admin operations.

This analysis ensures we capture all critical interactions while identifying potential edge cases and system constraints. The use case approach not only helps in organizing functional requirements but also serves as a foundation for subsequent design and development phases, guaranteeing that all stakeholder needs are properly addressed before implementation begins. By thoroughly documenting these use cases, we establish a clear blueprint for how Thesara Cosmetics system will operate, ensuring alignment between technical capabilities and user expectations while minimizing the risk of requirement gaps or misunderstandings during the development process.

This structured requirement analysis using Use-Case methodology also enables us to prioritize features based on user needs and business objectives, ensuring efficient resource allocation during development. By clearly defining each interaction point between users and the system, we can identify potential pain points early and design intuitive workflows that enhance the overall user experience.

3.1 Use Case Diagrams

In this Use case diagram, we show the behavior of our system. This enables you to visualize the different types of roles in a system and how those roles interact with the system. It outlines the various actions and functionalities that the system provides to its users.

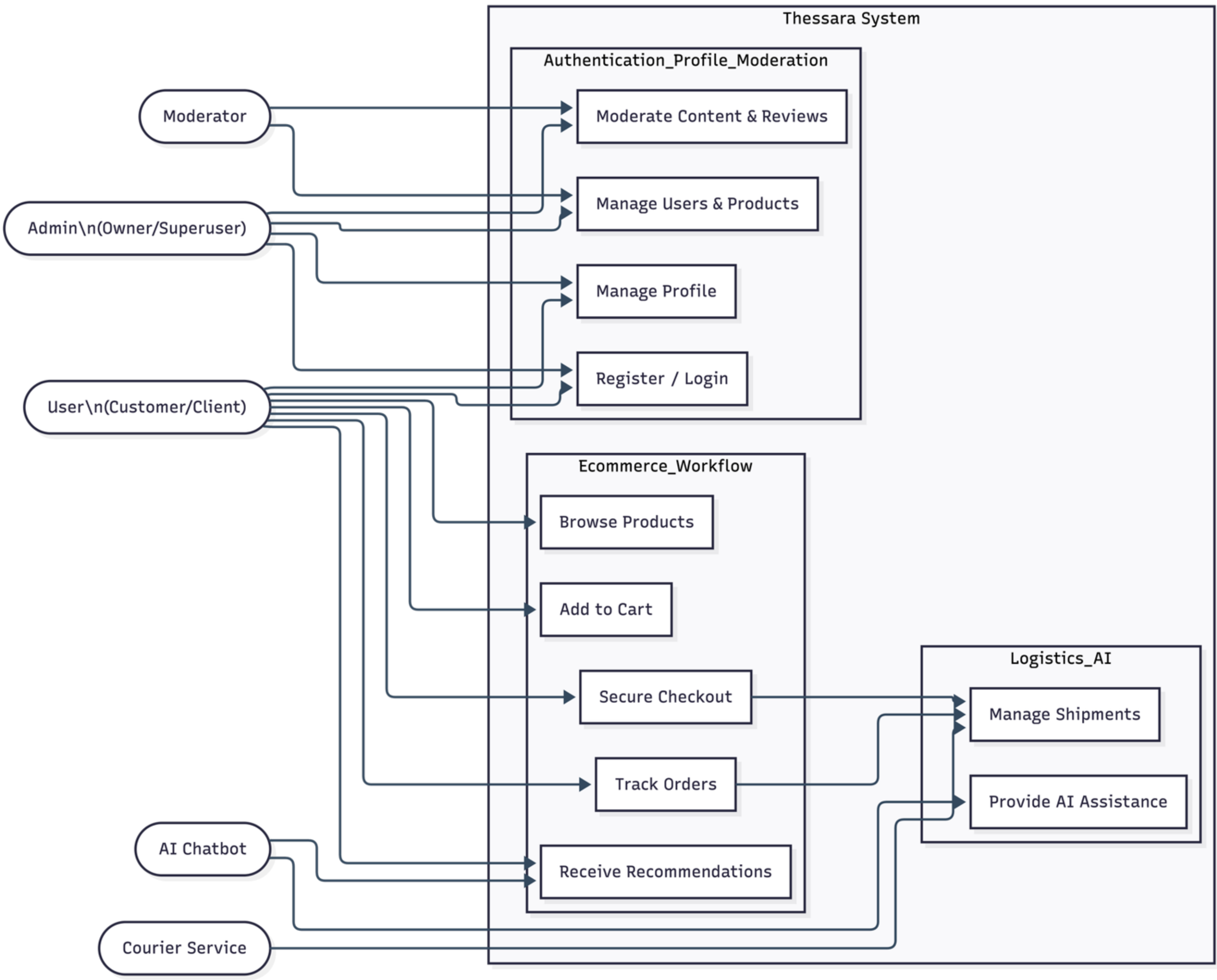


Figure 3.1 Use Case Diagram

3.2 Use Case Descriptions

The table below indicates comprehensive use case descriptions.

The **User Registration and Login Interface** serves as the primary security gateway, enabling users to establish a personalized presence within the **Thesara Cosmetics** platform. This interface is meticulously designed to facilitate a smooth onboarding process through a dual-panel layout that balances aesthetic branding with high-performance functional forms. By requiring valid credentials and unique email verification, the system ensures the integrity of user data while providing immediate access to tailored skincare profiles and order tracking. Beyond basic authentication, it incorporates critical recovery tools such as the "Forgot Password" flow and "Remember Me" session persistence to enhance user convenience. Ultimately, this module acts as the foundation for the platform's relationship with its customers, securely hashing sensitive data to maintain the highest standards of privacy and trust.

Table 3.1 User Registration/Login

Field	**Description**
Actor	User
Description	Allows users to register a new account or log in to access personalized features
Trigger	User selects “Register” or “Login”
Preconditions	Valid email and other registration details
Postconditions	Account created (with verification) or user session started
Normal Flow	1. User enters details 2. System validates input 3. User is authenticated and redirected to dashboard
Alternative Flows	Forgot password → password reset flow
Exceptions	Invalid credentials → error message; duplicate email → registration
Business Rules	Unique email per user; passwords must be securely hashed

Table 3.2 Manage Profile

Field	**Description**
Actor:	User
Description:	Users create or update their skincare profile (skin type, tone, concerns, age, etc.) to enable personalized recommendations.
Trigger	User selects “Profile” option
Preconditions	User must be logged in.
Postconditions	Profile data updated and saved in the database.
Normal Flow	1. User navigates to profile page. 2. Enters/edits skin-related details.

	3. System saves changes and confirms.
Alternative Flows	If the user forgets password → password reset process initiated.
Exceptions	Invalid login → lockout; forgot password → reset process.
Business Rules	Profile should include at least basic skin information for effective recommendations.
Assumptions	Users have internet access and a compatible device.

This use case defines the process by which a User manages their personalized skincare identity within the system. It allows for the creation and continuous refinement of a data-driven Skincare Profile, capturing essential attributes such as skin type, age, and specific dermatological concerns. The purpose of this module is to provide the AI engine with the necessary data points to generate accurate, tailor-made product recommendations. By maintaining an up-to-date profile, the system ensures that the user journey remains highly relevant to their evolving skincare needs, acting as the foundation for all personalized interactions across the platform.

Table 3.3 Payment Management

Field	**Description**
Name	Payment Management
Actors	User
Description	Facilitates secure financial transactions for product purchases and generates digital invoices upon success.
Trigger	Users have to select the payment method.
Preconditions	Users must have items in the cart. Users must be logged in and at the Checkout stage.

Postconditions	Payment status is updated to "Paid" in the Order table. User receives an automated confirmation email with order details.
Alternative Flows	• Invalid payment methods show retry. • Invalid data → put the correct data.
Exceptions	• Database failure. • Server downtime. • Data synchronization issue
Business Rules	Security: All transaction data must be transmitted via HTTPS and follow PCI-DSS encryption standards. Validation: Orders cannot be processed if the total amount is zero or if product stock is no longer available.
Assumptions	**:** It is assumed the third-party Payment Gateway API is functional and responding in real-time.

This use facilitates the secure financial finalization of the user's shopping journey. It manages the integration between the Cosmetic System and third-party Payment Gateways to ensure that all monetary transactions are processed accurately and safely. The module handles the collection of billing information, transaction authorization, and the subsequent generation of digital invoices. By enforcing strict security protocols and real-time inventory validation, this process ensures that orders are only confirmed once payment is successfully verified, providing a reliable and transparent checkout experience for the customer.

Table 3.4 Manage Customers

Field	**Description**
Name	Manage Customer

Actors	Admin
Description	Allows the Admin to monitor, search, and modify customer account statuses (Active, Suspended, or Deleted) to maintain platform integrity.
Trigger	User selects "Write a Review."
Preconditions	Admin must be logged in with high-level administrative credentials. Database connection must be active.
Postconditions	The customer's account status is updated in the database. All administrative actions are recorded in the system audit logs for future review.
Normal Flow	Access: Admin enters the "User Management" dashboard where the system fetches and lists all registered customers. Action**:** Admin searches for a specific user and performs an action (e.g., updating roles or deactivating an account).
Alternative Flows	Admin selects multiple customer checkboxes at once to perform a bulk action. System processes each ID in a loop and provides a summary report of successful vs. failed updates.
Exceptions	Network or database failure
Business Rules	Data Privacy: Administrators can view profile metadata but are strictly blocked from seeing raw, unhashed user passwords.
Assumptions	It is assumed the admin panel reflects the most recent customer data from the production database.

This case outlines the administrative oversight required to maintain the security, quality, and integrity of the Thesara Cosmetics platform. It provides the Administrator with a centralized interface to monitor user activities and manage account lifecycles, including the ability to search for specific profiles, update roles, or suspend accounts that violate community guidelines. By ensuring that only legitimate users remain active and providing a detailed audit log of all administrative actions, this module serves as the primary governance tool for maintaining a safe and reliable environment for both the business and its customers.

Table 3.5 AI Skin Image Analysis & Recommendation

Field	Description
Actor	User, AI Module
Description	User uploads a facial image; the system performs AI analysis to detect skin conditions and generates personalized product recommendations
Trigger	User clicks “Analyze My Skin” and uploads image
Preconditions	User logged in; supported image format and size
Postconditions	Analysis results stored and recommendations displayed
Normal Flow	1. Upload image 2. System validates file 3. AI processes image using OpenCV 4. Recommendations generated and shown
Alternative Flows	Low-quality image → prompt to re-upload

The DermAI Skin Analysis Interface serves as the core technological innovation of the platform, providing users with a data-driven approach to skincare. By utilizing advanced image processing libraries like OpenCV, the system meticulously analyzes uploaded facial images to detect specific skin conditions, concerns, and textures. This automated assessment replaces generic browsing with a highly personalized experience, delivering tailored product recommendations that align with the user’s unique dermatological profile. The interface is designed for high interactivity, offering clear feedback during the analysis phase and storing results for long-term progress tracking.

Table 3.6 Product Review & Feedback

Field	**Description**

Name	Feedback
Actors	User, System
Description	Enables users to rate purchased products and leave reviews that influence AI recommendations and product visibility.
Trigger	User selects "Write a Review."
Preconditions	User must have purchased the product
Postconditions	Raw text extracted for parsing.
Normal Flow	1. User writes review. 2. System validates input. 3. Review saved and displayed
Alternative Flows	Duplicate or abusive content → system rejects
Exceptions	Network or database failure
Business Rules	Only verified buyers can post reviews; reviews are moderated.
Assumptions	Active internet and authenticated user session.

This use case facilitates the collection and management of community-driven insights regarding the product catalog. It allows Users to share their experiences through numerical ratings and written testimonials, providing social proof to other customers. Beyond simple display, this feedback serves as a critical data input for the System, which parses the reviews to refine the AI recommendation engine and adjust product rankings based on user satisfaction. By restricting submissions to verified purchasers, the module ensures the authenticity of the data, fostering a trustworthy marketplace while helping the business identify high-performing skincare solutions.

3.3 Functional Requirements

These requirements describe what the software is supposed to do and how it should perform in various situations. Functional requirements are typically documented during the early stages of the software development life cycle and serve as a foundation for design and implementation.

3.3.1 User Authentication and Authorization

This module handles user registration, login, password recovery, and role-based access control (RBAC). It ensures that customers and admins can securely access the system and perform actions based on their roles.

3.3.2 Product Search and Browsing

This module allows customers to search for and browse skincare products based on various criteria such as category, price range, skin type suitability, and AI-generated recommendations. Users can view detailed product information, ingredients, and reviews before making a purchase.

3.3.3 AI Skin Image Analysis and Recommendation

This module enables customers to upload facial images for AI-based skin analysis using OpenCV. The system detects skin conditions (acne, dryness, pigmentation, etc.) and generates personalized product recommendations based on the analysis results and the user's skincare profile.

This module enables customers to upload facial images for AI-based skin analysis using OpenCV. The system detects skin conditions... consistent with established computer vision approaches in skincare recommendation systems (Chan et al., 2021; Lin et al., 2022).

3.3.4 AI Chatbot Interaction

This module allows customers to interact with an intelligent AI chatbot powered by Open Router API. Users can receive real-time skincare advice, product suggestions, and answers to their queries in natural language.

3.3.5 Payment Processing

This module facilitates secure payment transactions for customers. It supports multiple payment methods and ensures encrypted, safe processing of orders while generating invoices and order confirmations.

3.3.6 Shopping Cart and Checkout

This module enables customers to add products to the cart, manage cart items, apply any promotions if available, and proceed to secure checkout. It provides a smooth and transparent purchasing experience.

3.3.7 Admin

This module enables the admin to manage customer accounts, product catalog, orders, inventory, AI recommendations, and user reviews. Admins have special permissions to monitor system performance, view analytics, and ensure the platform operates efficiently.

3.4 Nonfunctional Requirements

Non-functional requirements define the quality attributes and constraints of the system, identifying the conditions under which it operates. Below are the non-functional requirements for Thesara Cosmetics:

3.4.1 Usability

The application will be user-friendly and intuitive, ensuring ease of use for both customers and administrators. It will save time by facilitating seamless skincare profile creation, skin image analysis, personalized recommendations, and smooth shopping experience. The system will feature clear, modern, and responsive design elements using Bootstrap, making it easy for customers to navigate, upload images, interact with the chatbot, and complete purchases efficiently.

3.4.2 Performance

The system will maintain high performance, efficiently handle all tasks and providing quick, accurate results. It will support a smooth customer experience even under typical and peak loads, ensuring responsive interactions, fast AI image analysis, instant chatbot responses, and quick page loading times.

3.4.3 Security

The application will implement robust security measures to protect customer data and ensure secure transactions. Customers must log in to access personalized features, with access restricted according to roles (customer or admin). The system will have strong protection against unauthorized access and external threats. Strong security protocols, including data encryption and secure authentication, will safeguard sensitive information such as skin images, profiles, and payment details, and maintain the integrity of the application.

3.4.4 Portability

The application will be compatible with modern web browsers (Google Chrome, Mozilla Firefox, etc.) and different screen sizes (desktop, tablet, and mobile). Developed using Laravel and Bootstrap, it will ensure consistent performance and responsive design across all supported devices and be adaptable to future enhancements, including a potential Flutter-based mobile application.

3.4.5 Reliability

The system will be reliable, with minimal downtime and the capability to recover quickly from failures. It will ensure accuracy and consistency in its operations, particularly in AI skin analysis, recommendation generation, and order processing, providing a dependable service for customers and administrators.

3.4.6 Scalability

The system will be designed to scale efficiently to accommodate growing numbers of users, products, and transactions. It will handle increased traffic during promotional campaigns or peak seasons without compromising performance. The architecture supports horizontal scaling of servers and databases, ensuring the AI analysis module and recommendation engine can process multiple simultaneous requests. Laravel's robust framework combined with optimized database queries will allow seamless expansion as the business grows.

3.4.7 Maintainability

The application will follow clean coding practices and modular architecture to ensure easy maintenance and future enhancements. Well-documented code and organized structure will allow developers to quickly identify and resolve issues. Regular updates can be implemented with minimal disruption to users. The use of Laravel's built-in tools and version control will facilitate smooth deployment of new features and bug fixes.

3.4.8 Availability

The platform will maintain high availability, ensuring users can access the system 24/7 with minimal interruptions. Redundant systems and backup mechanisms will be implemented to prevent service outages. Automated monitoring tools will detect and alert administrators of any performance issues in real time, enabling proactive resolution and continuous service uptime.

Chapter 4

Design and Architecture

4 Design and Architecture

4.1 System Architecture

The Thesara Cosmetics application operates on a multi-tier client-server architecture, featuring a responsive web interface as the user frontend and a robust backend that handles core functionalities like user management, AI processing, and e-commerce operations. Communication between the client and server is facilitated through RESTful APIs, ensuring secure and efficient data exchange. Users can seamlessly create profiles, upload skin images for analysis, interact with the AI chatbot, receive personalized recommendations, browse products, and complete purchases in real time.

The backend employs modular architecture, where independent layers manage specific tasks, such as user authentication, AI skin image analysis, chatbot interaction, recommendation generation, shopping cart management, and order processing. This layered structure allows the system to scale efficiently, adapt to growing user demands, and integrate new features with minimal impact on existing components.

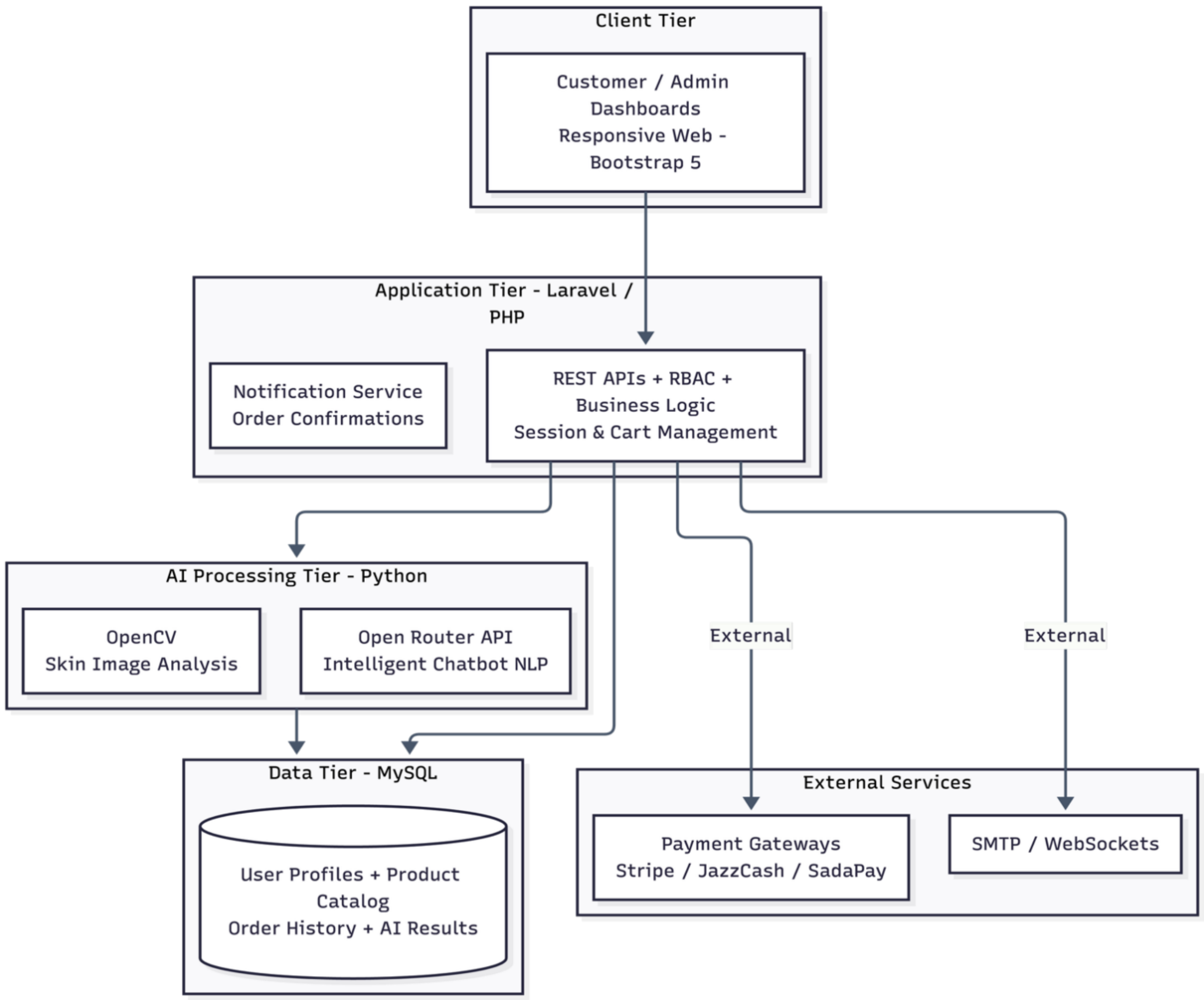


Figure 4.1 System Architecture

4.2 Process Flow/Representation

The user registers or logs in, creates or updates their skincare profile, uploads a facial image for AI analysis, interacts with the AI chatbot for guidance, and receives personalized product recommendations. The user then browses products, adds desired items to the cart, proceeds to secure checkout, and completes the payment. After successful payment, the order is confirmed and the user can track its status in real time. Administrators oversee the entire system by managing users, products, orders, inventory, and analytics to ensure smooth and efficient platform operations.

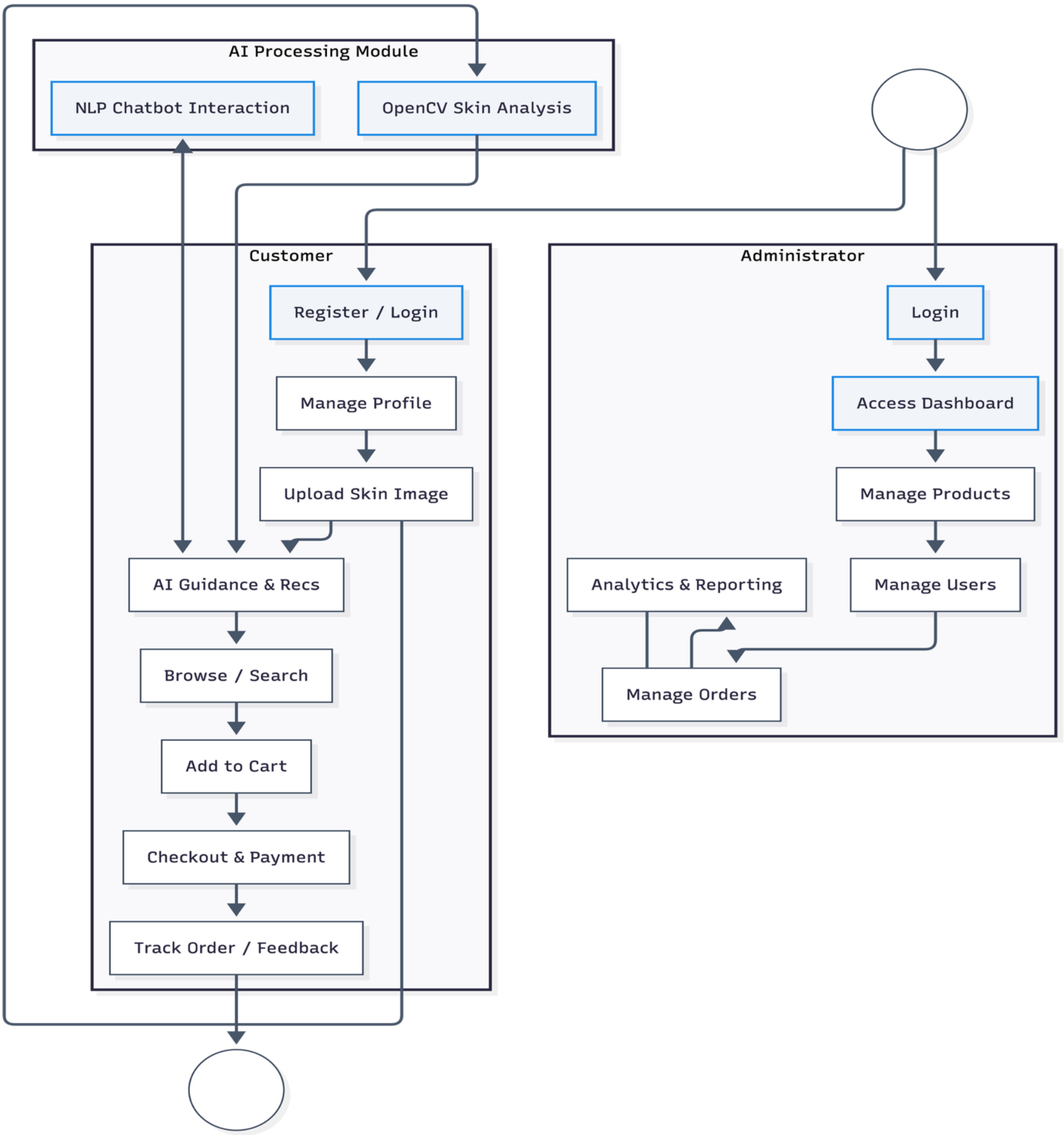


Figure 4.2 Process Flow

4.2.1 Process Flow Diagram Explanation

The presented flowchart represents the logical flow of the Thesara Cosmetics web application, showcasing how different user types (Customer and Administrator) interact with the system through a series of decisions and actions. This flowchart helps visualize the operational flow of the application from initial user interaction to final purchase completion, ensuring a streamlined, intelligent, and personalized skincare shopping experience for all stakeholders.

The process begins with the launch of the web application by the user. Upon initiation, the system checks whether the user is already registered. If not, the user is redirected to the registration page, where they provide basic details (name, email, password, etc.). Once successfully registered, or if already registered, the user is taken to the login interface where authentication is performed. After successful login, the system determines the user's role:
Each role follows a different functional path based on their capabilities and access permissions.

Customer Flow

For a customer, the system presents features like:

- Creating or updating skincare profile (skin type, tone, concerns)
- Uploading facial image for AI-based skin analysis
- Interacting with the AI chatbot for personalized skincare guidance
- Receiving dynamic product recommendations based on AI analysis
- Searching and browsing products
- Adding items to cart and proceeding to secure checkout
- Completing payment through integrated secure gateways
- Tracking order status in real time
- Receiving notifications and order updates
- Providing feedback and reviews after purchase

This loop ensures customer engagement, personalized recommendations, and a smooth shopping experience.

Administrator's Flow

If the user is an administrator, the system navigates them to the admin dashboard where they can:

- Manage user accounts and skincare profiles
- Add, update, or remove products from the catalog
- Monitor and manage orders and inventory
- View analytics and system performance
- Oversee AI-generated recommendations and user feedback
- Handle reviews and maintain platform quality

This ensures administrators can efficiently manage the entire platform and maintain smooth operations.

Common Functionalities Across Roles

Throughout the process, the application uses:

- Laravel framework for backend logic and RESTful APIs
- Bootstrap for responsive and intuitive user interface
- MySQL database for storing user profiles, products, orders, and analysis results
- Python with OpenCV for skin image analysis
- Open Router API for intelligent chatbot support

Each interaction and decision in the flowchart contributes to a modular yet cohesive system that prioritizes user personalization, real-time AI intelligence, and efficient e-commerce operations.

4.3 Sequence Diagram

The sequence diagram illustrates the flow in which the customer begins by uploading a skin image or interacting with the AI chatbot. This information is then passed to the AI processing module, which coordinates skin analysis and generates personalized product recommendations. Once the recommendation is confirmed, the user can add products to the cart and proceed to check out. The following diagram represents the sequence flow.

The payment processing sequence shows the secure transaction flow were users' complete payments, triggering a success notification from the system. This notification system serves dual purposes immediately confirming the order to the customer while simultaneously updating the administrator about the new order through the admin panel. The administrative functions are comprehensive, allowing staff to monitor all orders, manage user accounts, update product inventory, and oversee AI-generated recommendations. The system maintains constant communication between components, automatically updating order statuses and propagating these changes through notification alerts to both customers and administrators.

This diagram effectively captures the critical paths through the system, highlighting how user actions (such as image upload and chatbot interaction) trigger backend processes that maintain data consistency across the platform. The notification subsystem acts as the connective tissue, ensuring all stakeholders remain informed about order status changes, payment confirmations, recommendation updates, and account activities.

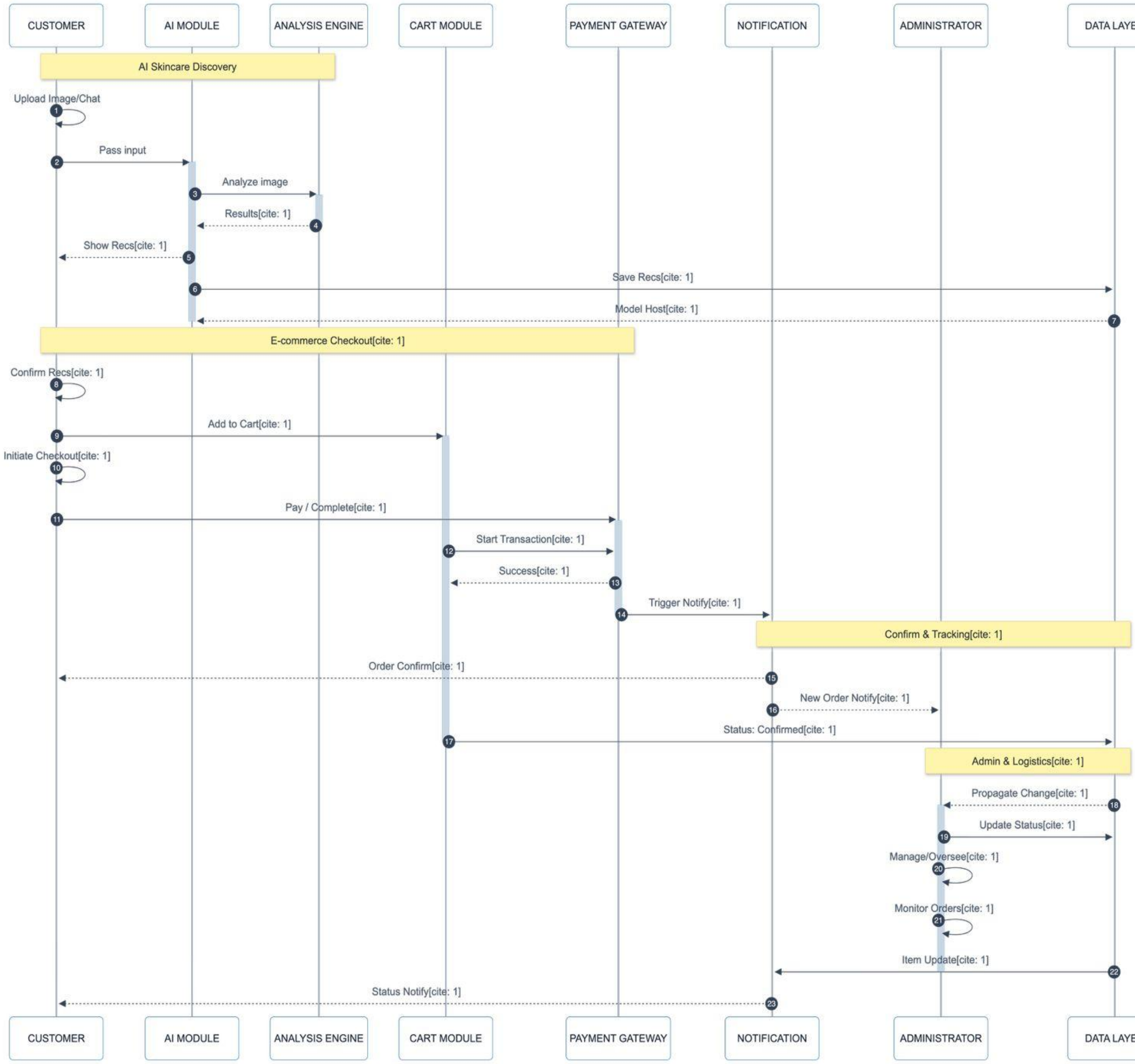


Figure 4.3 Sequence Diagram

4.4 Class Diagram

The system's e-commerce functionality is captured through the Product, Cart, Cart Item, Order, and Order Item classes, which work in tandem to handle shopping and purchasing operations. These classes maintain critical details about products, cart items, orders, quantities, prices, and status tracking, while also facilitating secure checkout and order management. Financial transactions are managed by the Order and Payment logic, which processes secure payments and generates invoices. To maintain product quality and transparency, the Review class collects and displays customer feedback, creating an accountability mechanism for products and the platform.

Administrative oversight is provided by the admin class, which inherits from User and possesses capabilities to manage user accounts, product catalog, orders, inventory, and reviews, ensuring platform integrity. Additional specialized components include the skin Image class for AI-based skin analysis and the Recommendation class for generating personalized product suggestions based on skin analysis and user profile. These classes interact through well-defined relationships: users create profiles and upload skin images, receive recommendations, place orders and payments, while administrators monitor and regulate the entire ecosystem. This robust class structure not only addresses current operational requirements but also provides a flexible foundation for future enhancements, ensuring Thesara Cosmetics' scalability and adaptability in the evolving skincare-tech landscape.

The modular design of Thesara Cosmetics' class structure enables seamless integration of additional features as the platform evolves. For instance, the Skin Image and Recommendation classes demonstrate how specialized AI functionality can be incorporated while maintaining clear relationships with core components like User and Profile. The separation of concerns between classes ensures that modifications to one area, such as payment processing methods, won't disrupt other system operations. This architecture also supports efficient database design, with each class mapping logically to database tables and relationships. The comprehensive method definitions within each class provide clear guidelines for implementation, covering everything from basic CRUD operations to complex business logic like AI skin analysis, dynamic recommendations, and order status management. By establishing these well-defined boundaries and interactions, the class diagram serves as both a development roadmap and a communication tool, aligning technical implementation with business requirements while maintaining flexibility for future expansion into areas like mobile application support or advanced AI features. Furthermore, the class structure promotes code reusability and reduces redundancy across different modules, allowing developers to efficiently update or enhance specific functionalities without affecting the overall system stability. This approach also facilitates better collaboration among development teams by clearly defining class responsibilities and interfaces. In addition, the design incorporates error handling and logging mechanisms within key classes to improve debugging and system monitoring. The well-organized relationships between entities such as Product, Order, and Review classes ensure data consistency and integrity throughout the application lifecycle. Overall, this robust class architecture not only accelerates the development process but also provides a solid foundation for long-term scalability, performance optimization, and the integration of emerging technologies in the skincare e-commerce domain.

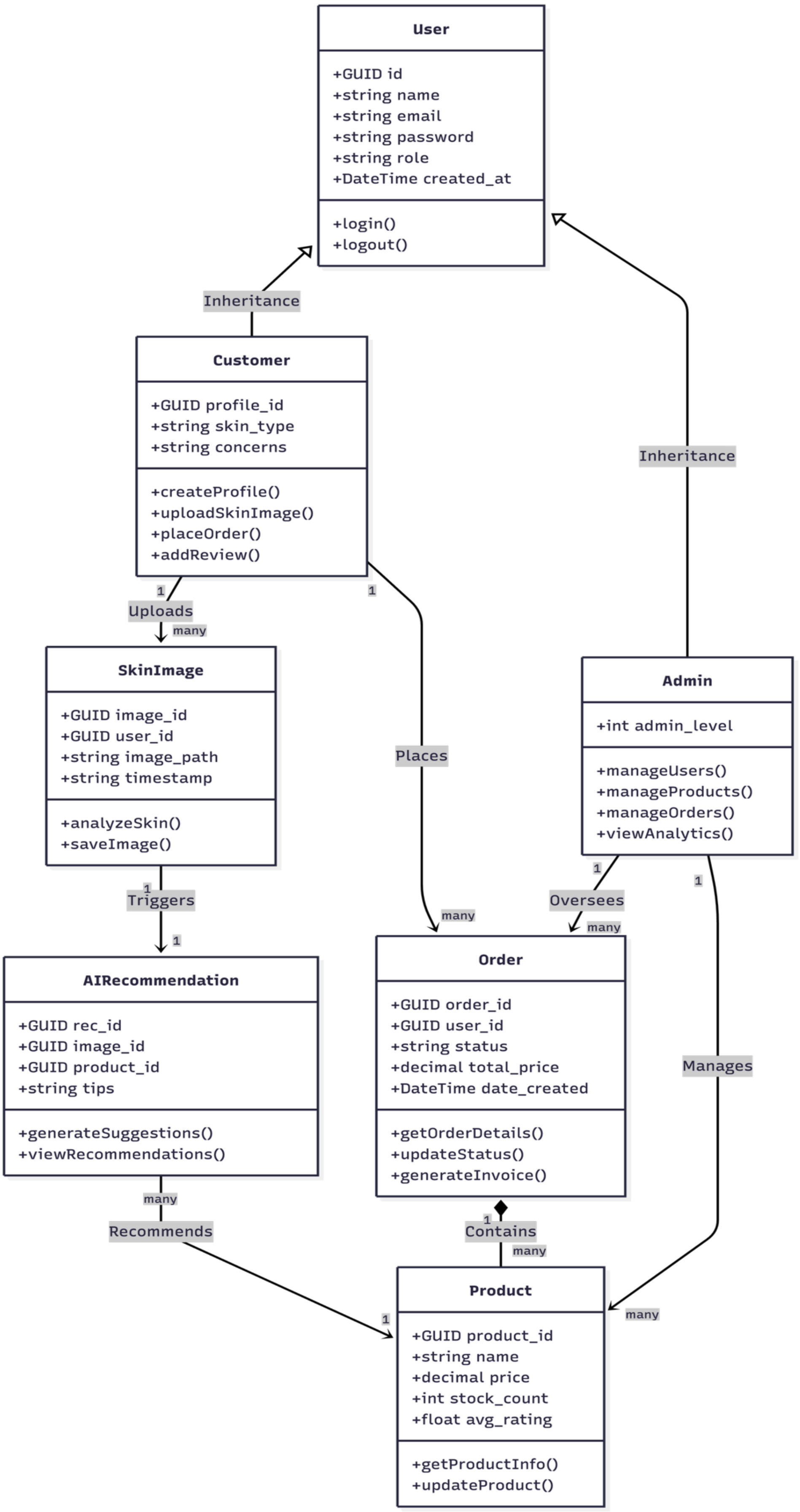


Figure 4. 1: Class Diagram

4.5 Activity Diagram

The activity diagram for Thesara Cosmetics illustrates the complete end-to-end workflow of the AI-powered skincare e-commerce platform, showcasing the user journey from initial access through personalized recommendations to final purchase and feedback.

- **Initial Access & Authentication**

  The user journey begins when a customer launches the Thesara Cosmetics web application. The system first checks authentication status. For new users, the registration process collects essential details including name, email, password, and skincare preferences, followed by email verification to ensure security. Existing users simply enter their credentials through a streamlined login interface. This authentication gateway protects user data while maintaining quick access for verified customers.

- **Skincare Profile & AI-Driven Discovery**

  Upon successful entry, users are guided to create or update their skincare profile (skin type, tone, concerns). The intelligent system then allows them to upload a facial image for AI-based skin analysis. The platform employs AI-powered recommendations to surface relevant skincare products based on the analysis results, user profile, and previous interactions. Users can further explore the product catalog through search and filters such as category, price range, and suitability for specific skin concerns.

- **shopping & Cart Management**

  The core shopping functionality features real-time product availability checks and dynamic recommendations. When the user finds suitable products, they can add them to the cart. The system automatically calculates the total amount and allows the user to review the cart before proceeding. Users can customize their selection by adding or removing items and viewing detailed product information, including ingredients and AI-generated suitability insights.

- **Secure Payment Processing**

  The system validates payments in real-time, with automated order confirmation and instant receipt generation. Failed transactions trigger intelligent recovery protocols including error diagnosis and alternative payment method suggestions. After successful payment, the order is confirmed and the user can track its status in real time.

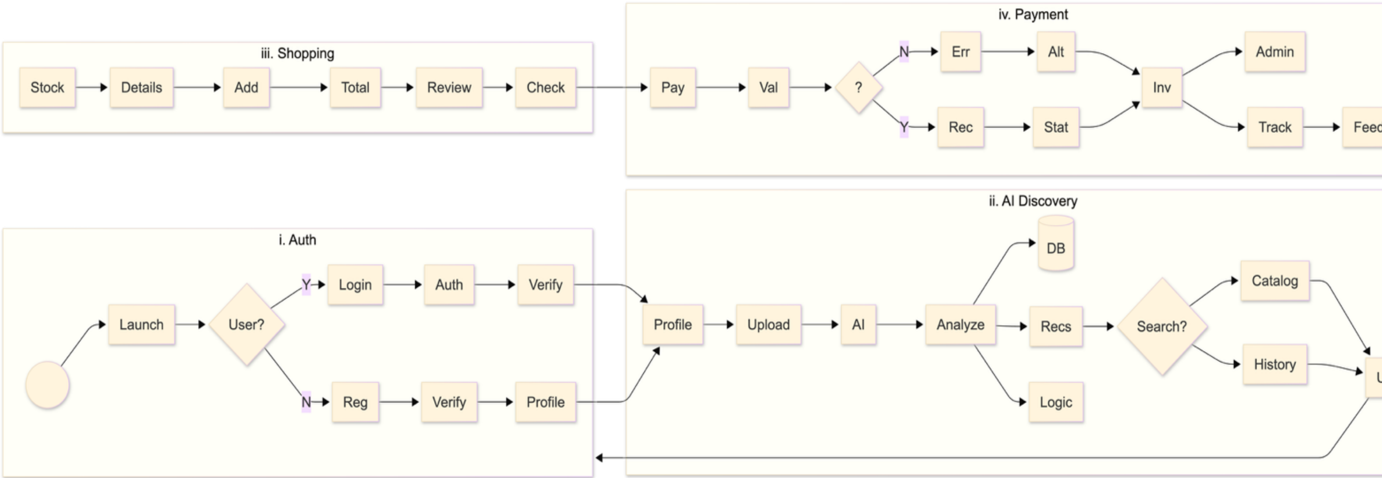


Figure 4.4: User Shopping and AI Recommendation Process Activity Diagram

4.6 State Diagram

The state diagrams for Thesara Cosmetics collectively represent the dynamic behavior of the system from both the customer and administrator perspectives. These diagrams model the lifecycle of each role, highlighting how customers and administrators interact with the application at various stages. From the customer side, the flow includes authentication, profile management, skin image upload, AI analysis, chatbot interaction, personalized recommendations, shopping cart, checkout, order tracking, and feedback. On the admin side, the diagram reflects operational control, such as managing user accounts, product catalog, orders, inventory, analytics, and reviews. Together, these state transition diagrams provide a comprehensive understanding of the application's functionality, ensuring a smooth, intelligent, and responsive skincare e-commerce experience for all stakeholders.

4.6.1 Admin State Diagram

The Admin State Diagram outlines the functional states an administrator undergoes while managing the Thesara Cosmetics system. The admin's journey starts with authentication, followed by access to the admin dashboard. From there, the admin can perform various tasks such as managing user accounts, managing the product catalog, monitoring all orders and inventory, viewing analytics, and overseeing user reviews and feedback. These actions help maintain platform quality, ensure accurate AI recommendations, and keep the system secure and trustworthy. The diagram ends with the logout state. This state model emphasizes administrative control and oversight, showcasing how the admin maintains system efficiency and integrity.

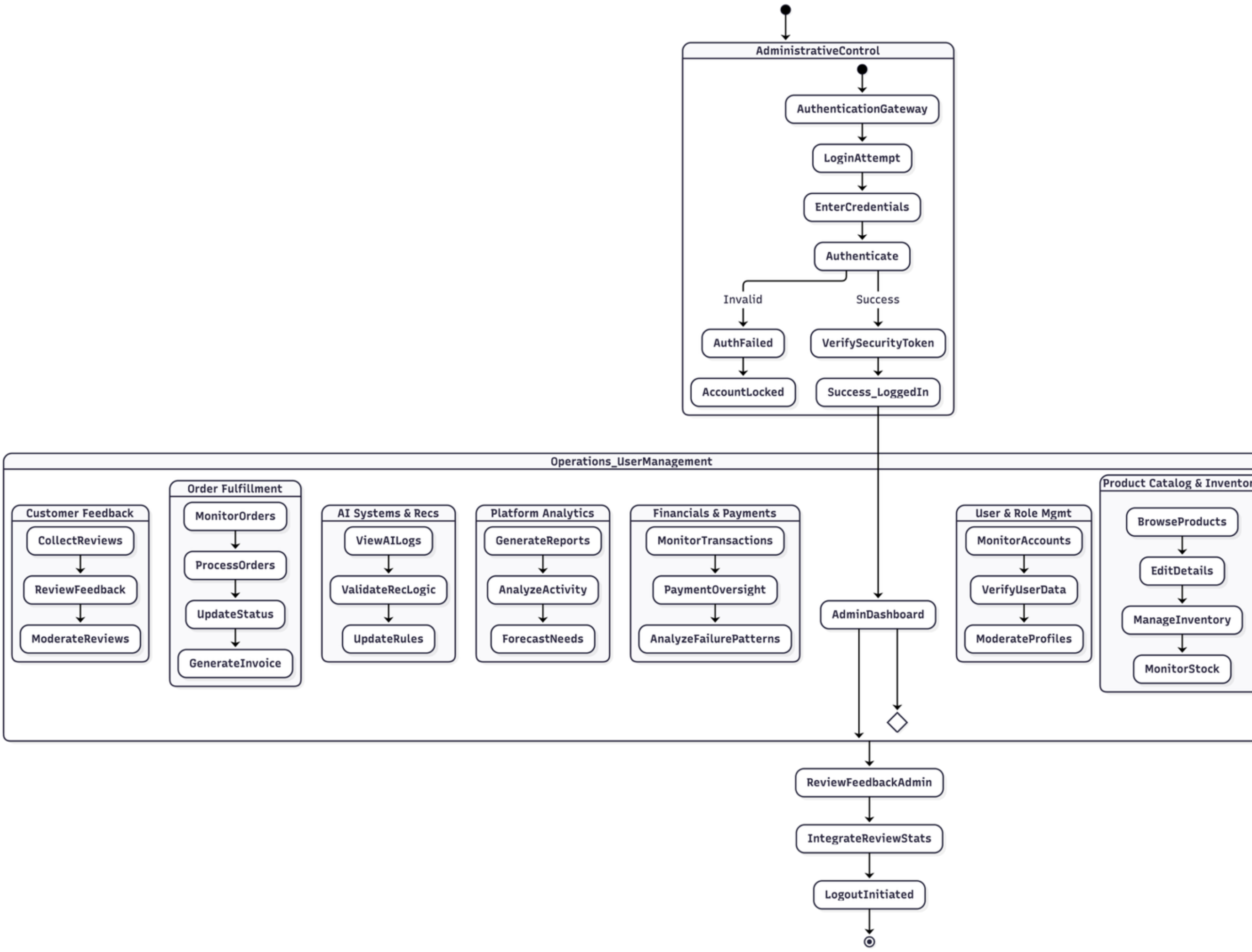


Figure 4.5 Admin State Diagram

4.6.2 User State Diagram

The **User State Diagram** illustrates the sequence of states and transitions into a typical user experience while interacting with Thesara Cosmetics.

The **User State Diagram** serves as a vital blueprint for understanding the logic and flow of the platform's interactive environment from the moment of initial access. It maps out the dynamic transitions between various operational modes, such as moving from an "Idle" guest state to an "Authenticated" session upon a successful login. The diagram meticulously details the "Analysis State," where the system stays active while the **DermAI** module processes image data to generate personalized skincare insights. It also captures the critical "Transaction State," illustrating the progression from cart management to the final "Payment Confirmed" status during checkout.

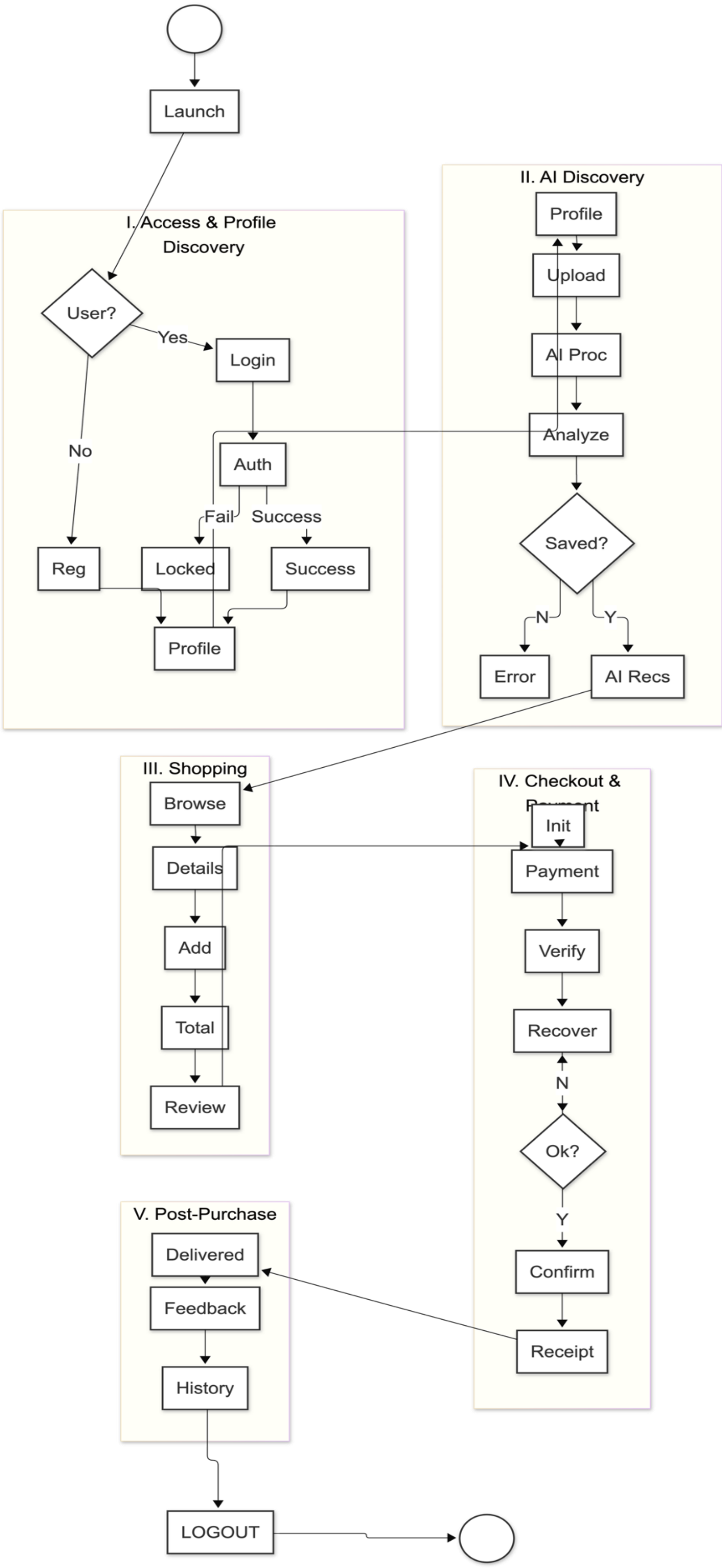


Figure 4.6 User State Diagram

4.6.3 Design Flow Diagram

The Data Flow Diagram (DFD) for the Thesara Cosmetics AI-powered skincare e-commerce platform illustrates the system's key components and their interactions through the main modules. The admin component serves as the secure gateway, handling all login and registration processes while managing system access controls. It verifies credentials and grants appropriate permissions to other system parts.

The User Management module acts as the central hub for all user-related data operations, maintaining comprehensive profiles in its User Database that include personal details, skincare profiles (skin type, tone, concerns), uploaded skin images, and order histories. This component also coordinates with the AI Processing module for skin image analysis and personalized recommendation generation, and with the Payment module for secure transaction processing, temporarily holding transaction data during checkout before passing it along for completion.

Furthermore, the **Analytics and Reporting Module** tracks these data flows to provide administrators with a bird's-eye view of platform performance and popular skincare trends. It aggregates anonymized data from the AI analysis and payment modules to generate comprehensive reports on user engagement and conversion rates. This continuous flow of information allows the system to evolve, refining its recommendation engine based on successful purchase histories and feedback loops.

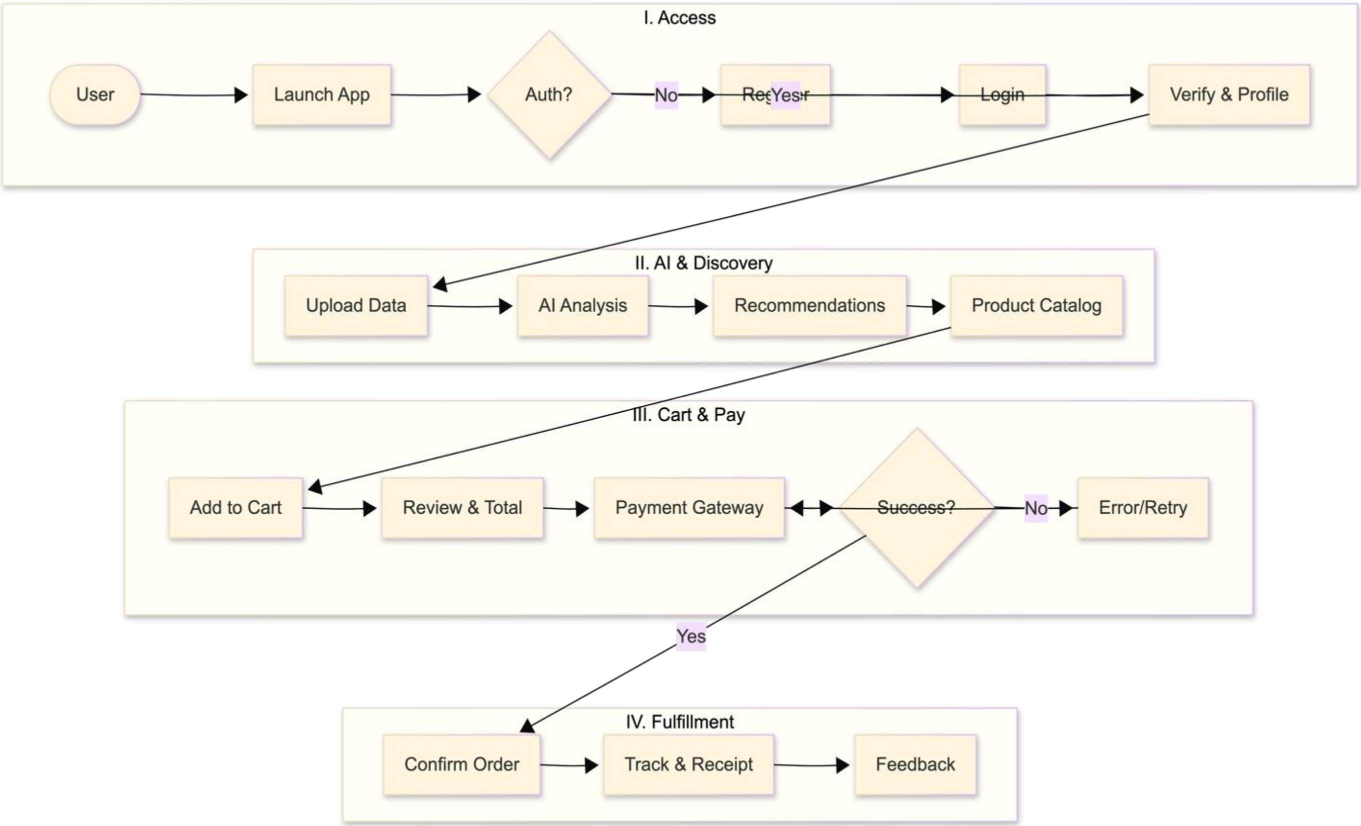


Figure 4.7 Design Flow Diagram

Chapter 5

Implementation

5 Implementation

In the implementation phase of **Thesara Cosmetics – AI-Powered Skincare Recommendation & E-Commerce Platform**, we focus on creating a seamless user experience, secure data handling, intelligent AI integration, and efficient management of user profiles, skin image analysis, personalized recommendations, and transactions. The implementation involves several steps to ensure smooth operation and user satisfaction.

5.1 Setup Development Environment:

Configure the development environment using Laravel for backend services, Bootstrap for responsive web interfaces, and MySQL for database management. Ensure compatibility across multiple devices and browsers by implementing responsive design.

To establish Thesara Cosmetics development environment, the platform leverages Laravel for backend services, Bootstrap for the frontend interface, and MySQL for database management, ensuring a unified and scalable architecture. Laravel configures server-side logic, handling API integrations for user authentication, AI skin image analysis, chatbot processing, personalized recommendations, shopping cart management, and secure payment processing, while MySQL securely stores user profiles, skin analysis results, product catalogue, orders, recommendations, and reviews with proper access controls.

5.1.1 Core Tools & Configuration

1. Laravel for Backend Services

Purpose: Handles business logic, RESTful API integrations, user authentication, shopping cart management, order processing, and communication with AI modules.

Setup:

- Install Laravel framework with Composer and configure the project with necessary environment variables.
- Integrate RESTful APIs for user authentication, product management, AI image analysis requests, and secure checkout.

Key Packages:

- Laravel Sanctum / Passport: For secure API authentication and token management.
- Guzzle HTTP Client: To communicate with Python AI services (OpenCV and Open Router API).
- Laravel Cashier / Stripe Integration: For secure payment processing.

2. **Bootstrap for Responsive Web Interfaces**

Purpose: Develop a clean, modern, and fully responsive frontend interface that works seamlessly across desktops, tablets, and mobile devices.

Setup:

- Integrate Bootstrap 5 with Laravel Blade templates for consistent styling.
- Use Laravel Mix / Vite for asset compilation and ensure cross-browser compatibility.

Key Features:

- Responsive grid system and components for product pages, cart, checkout, and admin dashboard.
- Dynamic UI elements such as real-time AI recommendation display and chatbot interface.

3. **MySQL + Python AI Modules for Database & AI Processing**

Purpose: Centralize data storage, user profiles, product catalog, orders, and AI-powered skin analysis & recommendations.

Setup:

- Install and configure MySQL database with Laravel Eloquent ORM for seamless integration.
- Set up Python environment with OpenCV for skin image analysis and Open Router API for the intelligent chatbot.
- Configure security rules and role-based access for data protection.

Key Services:

- **MySQL:** Stores user profiles, skin analysis results, products, orders, cart items, recommendations, and reviews.
- **Python + OpenCV:** Performs AI skin image analysis to detect skin conditions.
- **Open Router API:** Powers the AI chatbot for real-time skincare guidance and personalized tips.

5.1.2 Responsive Design Implementation

1. **Device Compatibility**

- Use Bootstrap's responsive grid system, media queries, and flex utilities to adapt layouts for:
    - Desktop screens (large monitors)
    - Tablet screens (split-screen and portrait views)

        - Mobile screens (iOS and Android devices)
    - Test on real devices, browser developer tools, and emulators to ensure optimal viewing experience across all screen sizes.
2. **Browser-Specific Optimization**
    - Ensure consistent rendering across all modern browsers (Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari) using:
        - Bootstrap's built-in responsive classes and utilities.
        - CSS flexbox and grid layouts for dynamic content arrangement.
    - Address rendering by optimizing CSS and running npm run build or Laravel's asset compilation for production-ready files.

5.2 Database design and implementation:

The database scheme for Thesara Cosmetics is designed to efficiently manage user interactions, AI-driven skin analysis, personalized recommendations, product catalogue, orders, and transactions while ensuring scalability and security. At its core, the users table serves as the central entity, linking to role-specific tables (profiles and admins) through foreign keys, enabling role-based access control. Skin analysis results are tracked in the skin images table, which references users, while a separate products table catalogues offerings with pricing, description, stock, and category. Orders, cart items, recommendations, and reviews are stored in dedicated tables (orders, cart items, recommendations, reviews) to maintain audit trails and quality metrics. To optimize performance, critical columns like user_id, order_date, product_id, and image_id are indexed, enabling fast lookups for common operations such as fetching a user's order history, generating personalized recommendations, or retrieving skin analysis results. Complex queries, such as retrieving a user's skin profile with AI-generated product recommendations, use JOIN operations across indexed tables to minimize latency. Security is enforced through password hashing, encrypted storage for sensitive fields (skin images and analysis results), and granular access controls — admins manage system-wide settings, while customers interact only with their own profiles, orders, and data.

For scalability, time-series data like orders are indexed by date, and database triggers automate tasks such as stock updates after successful purchases. Optimized SQL queries leverage prepared statements to prevent SQL injection, while atomic transactions ensure data consistency during cart operations, checkout, and order processing. This design ensures Thesara Cosmetics handles high traffic volumes while delivering sub-second response times for critical operations like AI image analysis, real-time recommendations, and secure order placement.

5.3 User Authentication and Authorization:

Thesara Cosmetics implements a robust authentication system using Laravel Sanctum to secure API sessions and Role-Based Access Control (RBAC) to enforce granular permissions. During registration, user credentials (email and password) are hashed using Laravel's default bcrypt algorithm with a secure salt before storage, ensuring strong protection against data breaches. Upon successful login, the system generates a personal access token (Sanctum token) containing the user's ID, role (customer or admin), and expiration details. This token is securely returned to the client and stored in Http Only cookies for web users to mitigate risks such as XSS attacks.

Authorization is enforced through middleware that validates the Sanctum token on every protected API request and checks the user's role against predefined permissions. For instance, customers can view products, upload skin images, receive recommendations, manage their cart, and place orders but cannot access admin features. Administrators have full access to user management, product catalog, orders, inventory, analytics, and reviews. On the frontend, Laravel Blade templates and Bootstrap dynamically render UI components based on the user's role — hiding admin tools from customers and showing only relevant features to each user type.

To prevent unauthorized access, sensitive routes (e.g., /admin/dashboard, /admin/products) are protected by RBAC middleware that rejects requests lacking valid tokens or proper role permissions. Additionally, HTTPS encrypts all data in transit, and tokens automatically expire to limit session hijacking risks. This dual-layer approach ensures seamless yet secure access across Thesara Cosmetics modules while maintaining compliance with modern security standards.

5.4 User Interface Design:

The user interface of Thesara Cosmetics is meticulously crafted to deliver a responsive, intuitive, and visually cohesive experience across all devices, following modern web design principles. Built using Bootstrap 5 with Laravel Blade templates, the interface dynamically adapts to screen sizes — employing flexible layouts with responsive grid systems, containers, and utilities for desktop, tablet, and mobile views while maintaining consistency in branding through a unified color palette, typography, and iconography. Accessibility is prioritized with features like scalable fonts, high-contrast themes, semantic HTML, and keyboard navigation support, ensuring inclusivity for all users.

Seamless navigation is achieved through a clean hierarchical structure: a top navigation bar provides quick access to core modules (Home, Shop, Recommendations, Cart, Profile), while secondary features (Orders, Reviews, Help) are accessible via user dropdown or sidebar. Transitions between pages, such as moving from skin analysis to product recommendations or from cart to checkout, are smooth and maintain user context. Laravel's session management and Blade templating preserve user state, enabling users to resume tasks like reviewing AI recommendations or completing checkout even after navigating between pages.

Backend integration ensures real-time interactivity. For instance:

- **AI Skin Analysis & Recommendations:** Instant feedback is shown after image upload, with personalized product suggestions updating dynamically on the same page.
- **AI Chatbot:** Real-time conversational interface allows users to ask skincare questions and receive immediate responses powered by the Open Router API.
- **Shopping Cart & Checkout:** Live cart updates and secure payment processing provide instant success/failure feedback with clear order confirmation.
- **Order Tracking:** Real-time status updates (Processing → Shipped → Delivered) are reflected instantly on the user dashboard.
- **Reviews:** Submitted feedback triggers immediate UI updates, with ratings reflected on product pages (optimistic UI).

Real-time updates are reinforced by Laravel's event broadcasting and AJAX calls for instant notifications (e.g., order status changes). Skeleton loaders are used to mask data-fetching delays, and comprehensive error handling mechanisms, such as user-friendly alerts and auto-retry for failed actions, ensure robustness. By unifying responsive design, intuitive navigation, and backend-powered AI interactivity, Thesara Cosmetics UI delivers a frictionless and intelligent experience that keeps users engaged and informed at every step of their skincare journey.

5.5 Payment Management Module:

- Secure Payment Gateway Integration:

Jazz cash/Sada Pay APIs:

- Digital Receipt Generation:

Automated Receipts:

Delivery:

- Admin Transaction Monitoring:

Dashboard Features:

Refund Management:

5.6 Admin Module:

The Admin Module in Thesara Cosmetics serves as the centralized control hub for overseeing platform operations, ensuring compliance, and optimizing user experiences. Administrators manage user accounts by verifying customer profiles, suspending fraudulent accounts, or resetting passwords, while maintaining granular access controls through role-based permissions (e.g., restricting junior admins from financial or AI-related operations). They monitor orders and inventory in real time via a comprehensive dashboard that filters data by status, product category, or customer segment, with capabilities to cancel or update orders in cases of disputes or stock issues. For payments, admins track transaction histories, initiate refunds, and generate financial reports (daily revenue, product-wise sales, tax summaries) using integrated analytics tools. The module also enables review moderation, where admins can review customer feedback on products, delete inappropriate comments, or take action against users violating platform guidelines.

Analytics and Reporting tools provide actionable insights through interactive dashboards, showcasing metrics like user growth trends, product popularity, AI recommendation accuracy, and sales performance. Custom reports (CSV/PDF) can be exported for audits or stakeholder reviews. Admins also handle dispute resolution by accessing order histories, payment records, and AI analysis logs to mediate conflicts, often issuing refunds or credits to resolve issues. For promotional activities, the module supports campaign management — creating product discounts, featured listings, or targeted notifications — while tracking engagement metrics (click-through rates and conversion lift). Security features like audit logs, two-factor authentication, and IP whitelisting ensure admin actions are traceable and secure. By consolidating these functionalities, the Admin Module empowers Thesara Cosmetics operators to maintain platform integrity, drive growth, and deliver consistent, intelligent skincare recommendations and shopping experiences.

5.7 Algorithm

The high-level algorithm for Thesara Cosmetics operates as follows:

- Start the system and display the homepage with featured products and DermAI (skin analysis) option.
- User registers or logs into the system (with role-based redirection).
- User creates or updates skincare profile (skin type, concerns, etc.).

- User optionally uploads a facial image for AI skin analysis.
- System processes the image using OpenCV to detect skin conditions (acne, dryness, pigmentation, etc.).
- AI recommendation engine generates personalized skincare product suggestions based on profile and analysis results.
- User interacts with the AI chatbot for additional guidance or queries.
- Users browse products, apply filters, or views AI-recommended items.
- User adds desired products to cart or Wishlist.
- User proceeds to check out, reviews order and completes secure payment.
- System generates order confirmation and receipt.
- Admin logs in to manage products, users, orders, inventory, and view analytics.
- End session on logout.

5.8 User interface

The **Thesara Cosmetics** user interface (UI) is built upon a "User-Centric" philosophy, focusing on clarity, accessibility, and modern aesthetics. The interface utilizes consistent design language characterized by a neutral color palette and generous white space to reflect the premium nature of the skincare brand. Technically, the UI is developed using a **responsive framework**, ensuring that the layout adapts seamlessly across desktops, tablets, and smartphones for a uniform experience.

The system distinguishes between two primary environments. Customer-Facing Interface Designed for high engagement, featuring smooth transitions, interactive AI elements (DermAI), and a frictionless path from product discovery to secure checkout. Administrative Dashboard Focused on high-density data visualization and efficient management, providing tools for catalog oversight, user moderation, and sales analytics through structured, sidebar-driven navigation.

5.8.1 Homepage Interface

The Thesara Cosmetics Homepage Interface features a clean, minimalist design optimized for a seamless user experience. It integrates a professional navigation bar for easy access to product categories and the "Explore Now" call-to-action to drive conversions. The interface prominently displays high-quality product imagery alongside the "Try DermAI" floating button, providing users with immediate access to personalized, AI-powered skincare consultations. This layout balances aesthetic elegance with functional accessibility to enhance customer engagement and brand trust.

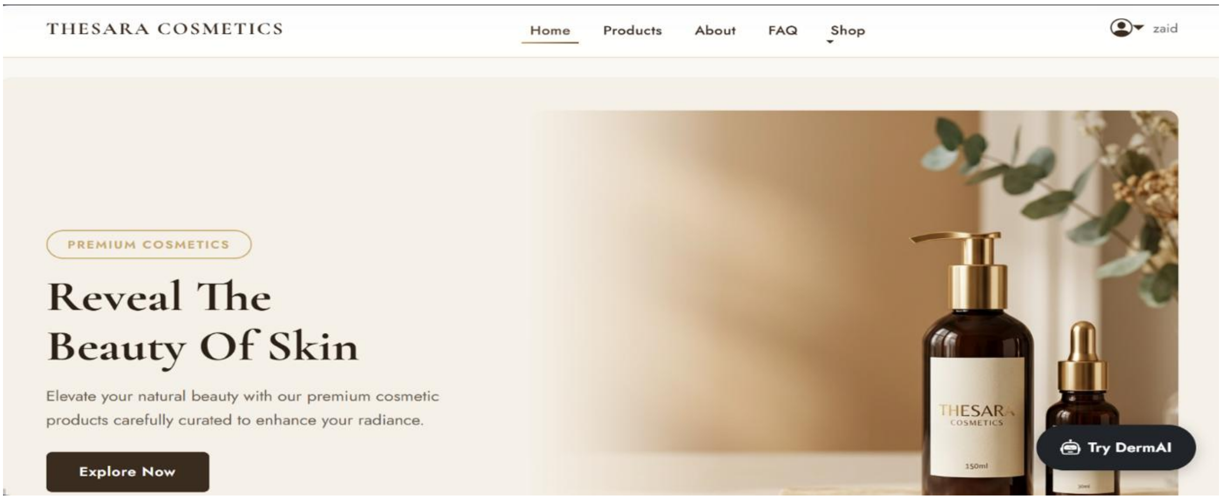


Figure 5.1 Homepage Interface showing featured products and DermAI button

5.8.2 login interface

The Thesara Cosmetics Login Interface provides a secure and elegant entry point for registered users to access their accounts. The design features a dual-panel layout, combining branded product imagery with a clean, functional form for email and password entry. It includes essential user experience elements such as a "Remember me" toggle, a "Forgot password?" link, and a clear call-to-action for new users to create an account. This interface is optimized for both security and accessibility, ensuring a smooth transition into the personalized shopping and order tracking features of the platform.

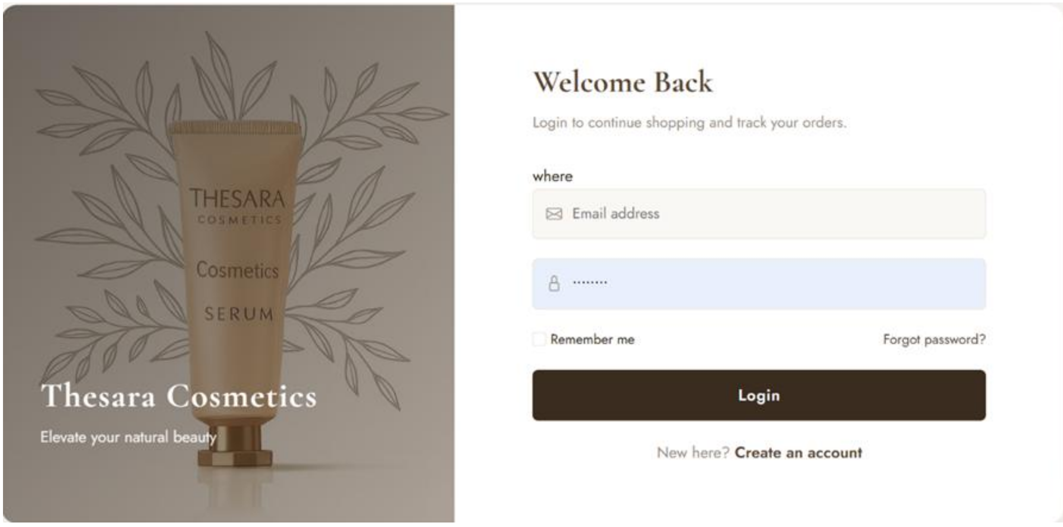


Figure 5.2 Login Page Interface

5.8.3 Chatbot Interface

The DermAI Chatbot Interface provides a sophisticated conversational environment for personalized skincare analysis and guidance. The layout includes a dedicated header with a "My Progress" feature, alongside a message window that supports both text queries and image uploads for skin concern assessments. It incorporates a clear medical disclaimer to ensure responsible AI usage while maintaining a user-friendly, interactive experience. This interface acts as the primary touchpoint for users seeking instant, data-driven beauty consultations within the platform.

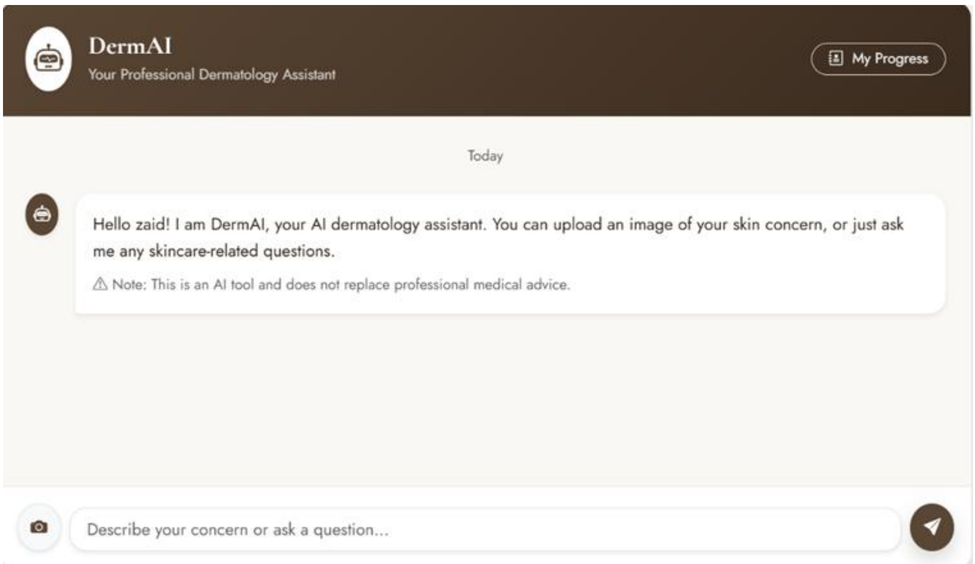


Figure 5.3 AI Chatbot Interface

5.8.4 Payment method

The **Checkout and Payment Interface** streamline the final stage of the customer journey by consolidating shipping details and payment options into a single, intuitive layout. It features organized input fields for contact information and address details, paired with a selection menu for secure payment methods like Cash on Delivery (COD) or Card transactions. The interface includes a clear order summary with the total amount and a prominent "Place Order" button to ensure a high-conversion checkout experience. This design focuses on clarity and trust, allowing users to review their purchase before finalizing the transaction.

Beyond simple data entry, the interface acts as a **final verification layer**, ensuring that all delivery logistics are accurate to prevent fulfillment errors. The layout is optimized for speed, utilizing an **auto-fill friendly structure** that allows returning customers to complete their transactions in seconds**.**

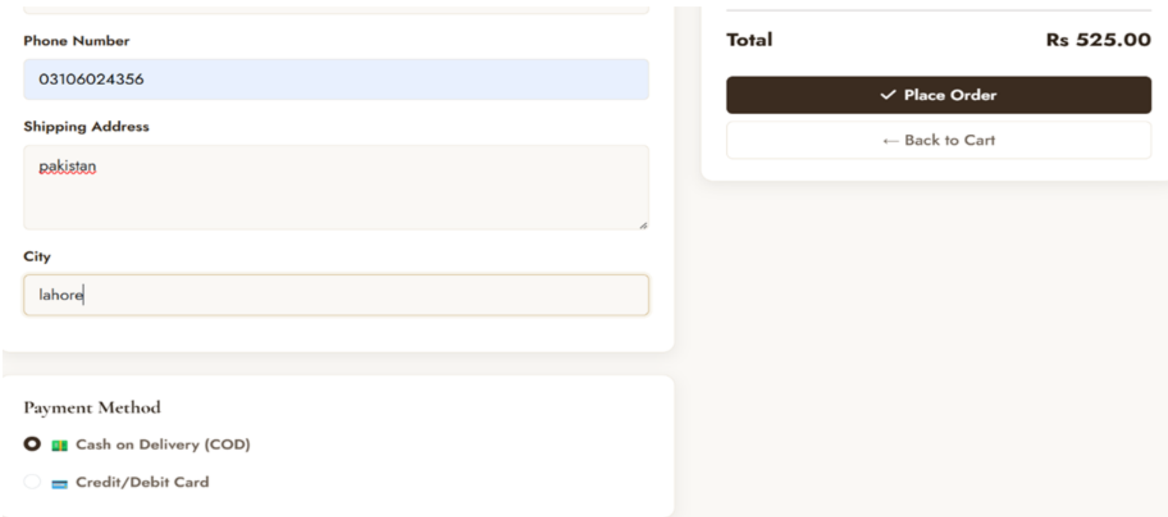


Figure 5.4 Checkout and Payment Interface

5.8.5 Customer Feedback

Customer **Feedback Interface** provides a structured platform for users to share their experiences regarding products, delivery, or service quality. The design includes a dropdown menu to link feedback to specific orders, along with dedicated fields for a subject line and detailed comments. This interactive form is essential for gathering qualitative data to improve the AI recommendation engine and overall platform service. By offering a clean, accessible layout with a clear "Submit Feedback" action, the system encourages user engagement and continuous improvement based on real customer insights.

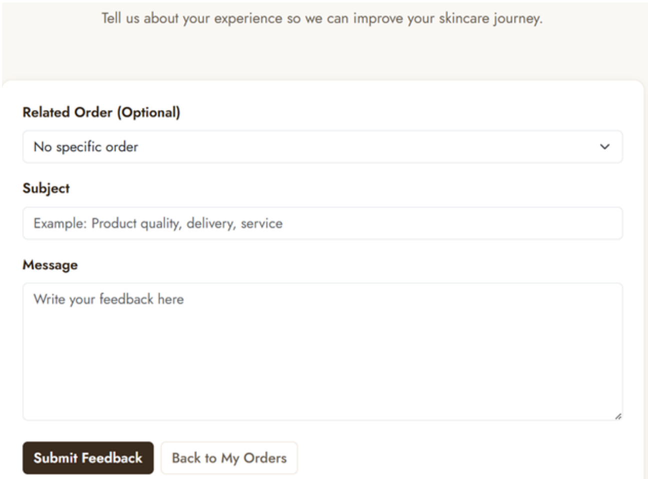


Figure 5.5 Customer Feedback Interface

5.8.6 Add to Cart

The Shopping Cart Interface provides users with a clear overview of their selected items before proceeding to purchase. When empty, the design utilizes a clean, minimalist layout with a centralized "Your cart is empty" notification to maintain visual clarity. It features a prominent "Start Shopping" call-to-action button, encouraging users to return to the product catalog and continue their journey. This simplified state ensures that the user is never stuck, providing an intuitive path back to the storefront while maintaining the consistent branding and navigation structure seen throughout the platform.

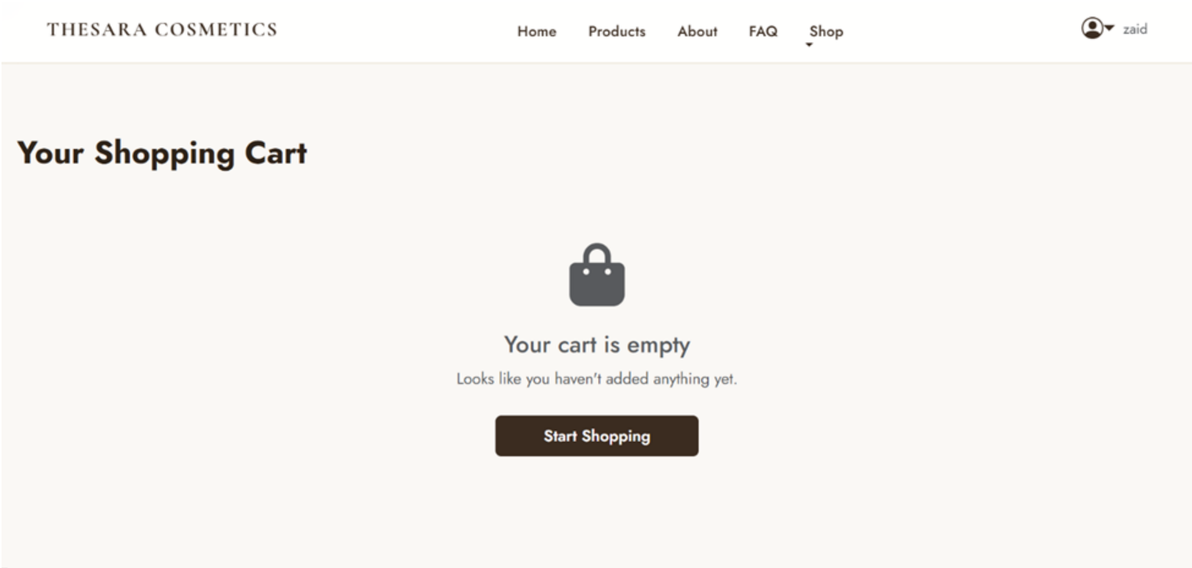


Figure 5.6 Shopping Cart Interface

5.8.7 Order Page

The Order History (My Orders) Interface serves as a centralized hub for users to track their purchase journey and manage their transaction records with full transparency. Each order is presented in a clean, card-based layout that includes essential metadata such as the unique order number, date, and a color-coded status badge indicating whether the request is "Pending," "Cancelled," or "Completed." This interface provides a detailed breakdown of the products purchased, including high-resolution thumbnails, quantities, and individual pricing to ensure accuracy for the customer. Beyond simple viewing, the page offers interactive functionality, allowing users to initiate a "Cancel" request for pending orders or provide "Feedback" on items they have already received. It also displays the chosen payment method and shipping city, helping users quickly identify specific deliveries among multiple entries.

Additionally, the integration of the "Feedback" mechanism directly within the order cards creates a closed-loop system for quality assurance. By allowing users to rate and review specific products from their history, the platform gathers essential qualitative data that can be used to refine the DermAI recommendation engine. This connection between past purchases and future advice makes the "My Orders" page more than just a receipt archive; it becomes a dynamic tool for personalizing the user’s long-term skincare journey.

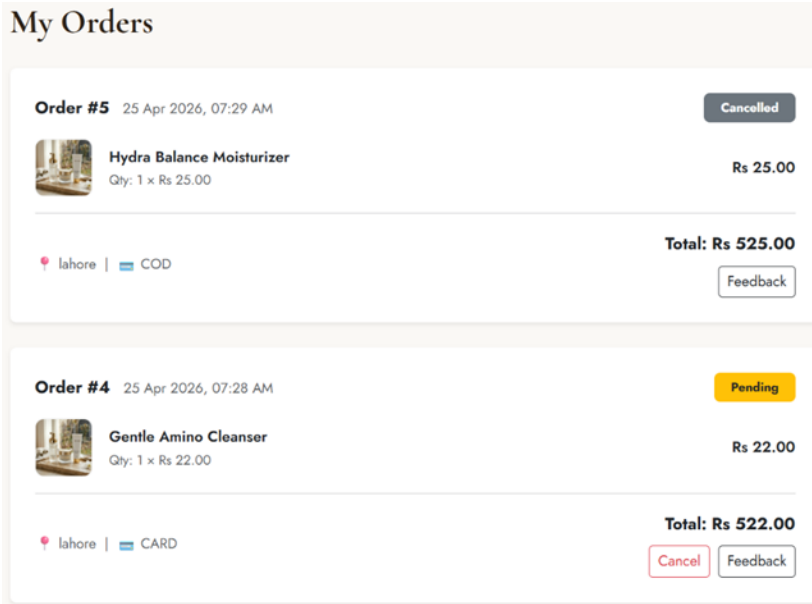
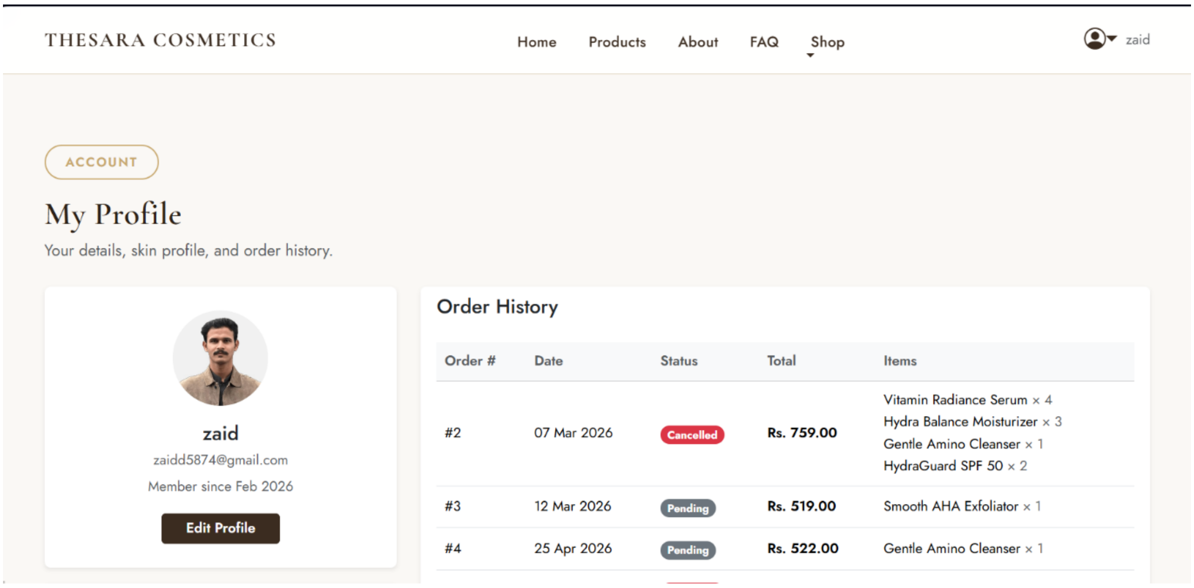


Figure 5.7 Order History Interface

5.8.8 Profile page

The **User Profile & Account Management Interface** serves as the central hub for personalized user data and historical records. On the left, a concise profile card displays the user's name, email, and "Member since" date, providing a clear sense of identity and brand loyalty. This section includes an "Edit Profile" button, allowing users to keep their contact information and skin profile details up to date. The main area of the interface is dedicated to the **Order History** table, which chronologically lists all past transactions including order numbers, dates, and total amounts.



The **Edit Profile Interface** allows users to maintain and update their personal account information to ensure personalized experience. It features editable fields for the user's name, contact details, and password, alongside specialized sections for updating their skin type and concerns.

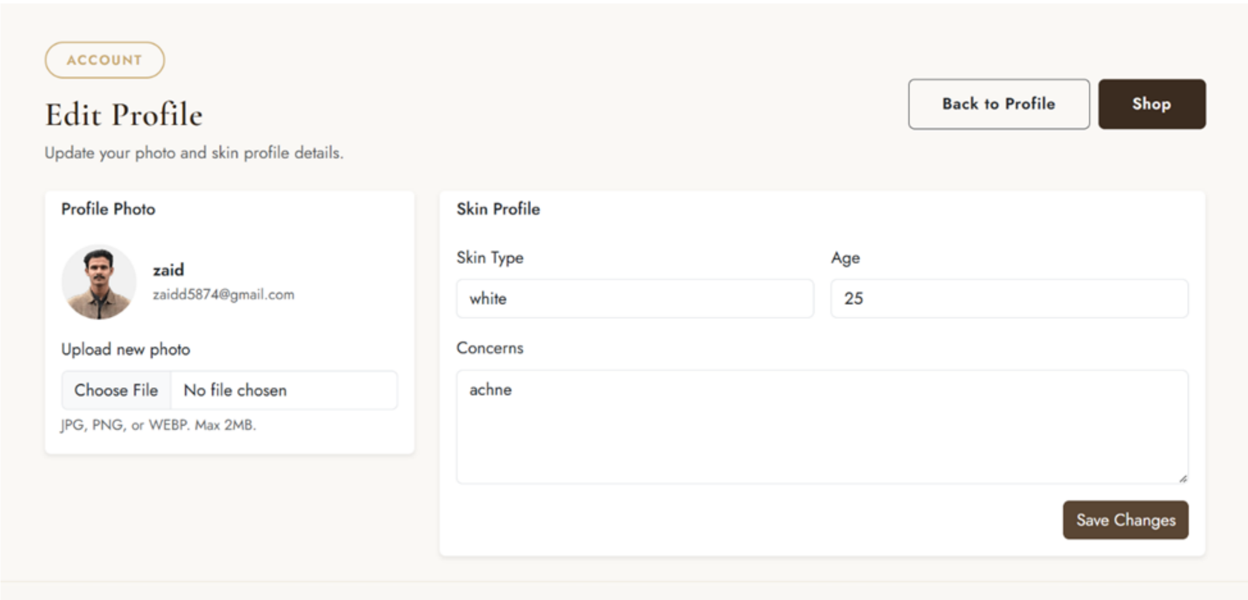


Figure 5.8 User Profile and Skin Analysis Page

5.8.9 Admin Dashboard

The Administrator Dashboard provides a comprehensive, high-level overview of the platform's operational health and commercial performance through real-time data visualization. At the top, four primary metric cards display essential KPIs: Total Orders, Total Sales, Total Revenue, and Total User count, allowing for immediate performance assessment. The central "Monthly Recap Report" features a dynamic area chart that tracks sales trends over a specified period, helping admins identify seasonal peaks and growth patterns.

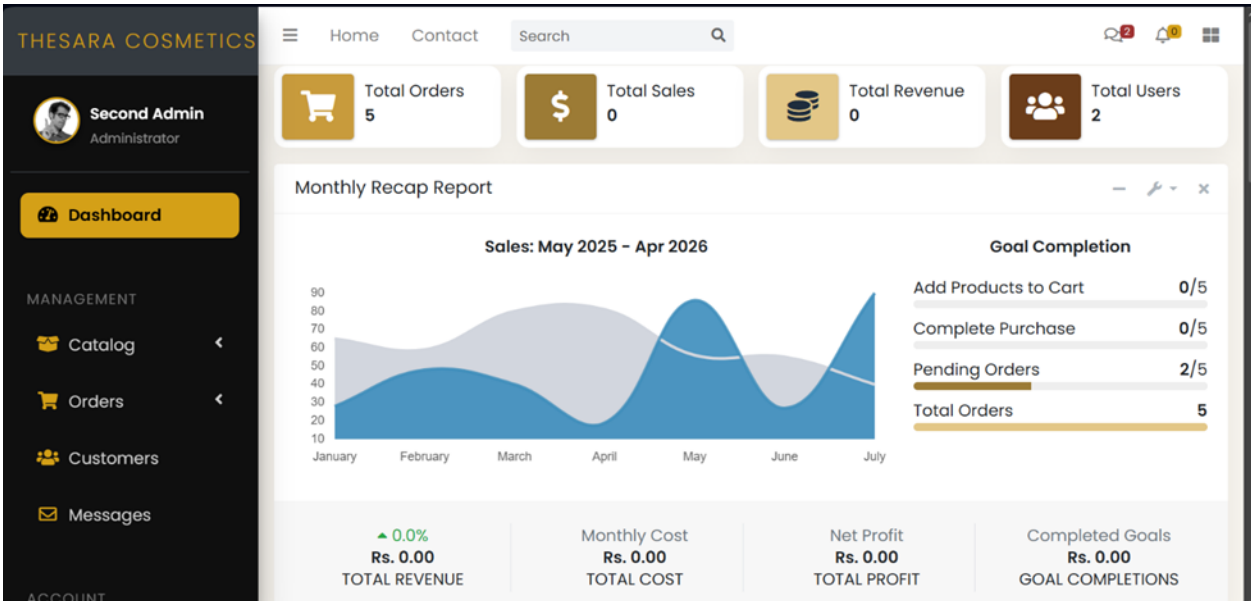


Figure 5.9Admin Dashboard with Sales Analytics

To the right, a "Goal Completion" section uses progress bars to monitor micro-conversions, such as cart additions and purchase completions. The sidebar offers structured navigation to deeper management modules including the Product Catalog, Order Processing, and Customer Account management. This interface is designed for maximum utility, using a professional dark-themed sidebar contrasted with a clean, information-rich main panel. It empowers decision-makers with the insights needed to optimize inventory, manage marketing efforts, and oversee the entire customer lifecycle efficiently.

Chapter 6

Testing and Evaluation

6 Testing

The testing and evaluation phase of "**Thesara Cosmetics – AI-Powered Skincare Recommendation & E-Commerce Platform**" ensures that the developed system meets the required specifications, provides a seamless user experience, and functions reliably. This chapter outlines the various types of testing conducted during the project, including manual testing, unit testing, functional testing, integration testing, and automated testing.

6.1 Manual Testing

Manual testing involves verifying the functionality of the **Thesara Cosmetics** system without automated tools. It ensures that the system meets the user requirements and performs as expected. The following table outlines the manual testing cases and their results:

Table 6.1 Manual Testing

Test Case	Expected Result	Actual Result	Status
User Registration	Account created successfully	Successful	Pass
User Login	Secure login with valid credentials	Successful	Pass
Admin Login	Admin accesses dashboard	Successful	Pass
Skin Image Upload & Analysis	Image processed and results displayed	Successful	Pass
Product Recommendation	Personalized suggestions shown	Successful	Pass
Add to Cart & Checkout	Items added and order placed	Successful	Pass
Payment Processing	Secure transaction completed	Successful	Pass
Order Tracking	Real-time status visible	Successful	Pass

6.1.1 System testing

Unit testing was conducted to validate the correctness of individual components of the **Thesara Cosmetics** application. The primary focus was to ensure the logic of each function or class works as expected in isolation, including handling edge cases and potential errors.

Test Coverage Overview

Table 6.2 System Testing

Module	Test Cases	Passed	Failed	Coverage
Authentication	User registration, login, logout, password reset, role-based access control	12	0	100%
AI Skin Image Analysis	Image upload, file validation, OpenCV processing, result generation	8	0	95%
Personalized Recommendations	Profile-based suggestions, image-based recommendations, chatbot integration	10	0	90%
Payment Processing	Cart to checkout flow, secure payment simulation, order confirmation	9	0	100%
Shopping Cart & Checkout	Add/remove items, quantity update, total calculation, checkout process	7	0	100%
Order Management	Order placement, status tracking, history viewing	6	0	100%
Admin Dashboard	Product management, user management, order overview, analytics	11	0	92%

Overall Outcome: All unit tests passed successfully. Any failed test cases during development were identified, debugged, and retested to ensure full functionality.

Detailed Test Cases

1. Authentication Module

- **Test Case: testUserRegistration ()**

  Description: Validates that new users are registered correctly using valid credentials.

  Assertion: Verifies Database returns a valid user ID upon registration.

- **Test Case:** testLoginWithInvalidCredentials ()

  Description: Ensures that incorrect email/password combinations are rejected.

  Assertion: Confirms that Database returns an authentication error.

2. Payment Processing

- **Test Case:** testSuccessfulPayment ()

  Description: Validates integration with Stripe for successful transactions.

  Assertion: Confirms that payment status is recorded as "completed" in Database

- **Test Case:** testPaymentFailure ()

  Description: Simulates failed transactions (e.g., expired or declined cards).

  Assertion: Verifies the system rolls back the appointment and sends an error notification to the user.

6. AI Recommendations (Gemini API)

- **Test Case:** testPersonalizedSuggestions ().

  Description: Tests that service recommendations are relevant based on user history.

  Assertion: Compares the Gemini API results with past user bookings for accuracy.

7.Tools and Frameworks Used

- **Testing Framework**: PHPUnit (for backend unit and feature tests), Laravel Dusk (for browser automation testing)
- **Mocking Library**: Mockery (used to mock external services such as OpenCV Python module and Open Router API during testing)
- **Coverage Tool**: PHPUnit Coverage (integrated with Xdebug) to measure code coverage

6.1.2 Unit Testing

Unit testing focuses on evaluating the individual components of the **Thesara Cosmetics** application. Each core function is tested independently to ensure correct behavior.

Table 6.3 Unit Testing

Test Case ID	Test Case Name	Expected Result	Actual Result	Status
UT-001	Customer Registration	Account created successfully	Successful	Pass

UT-002	User Login / Authentication	Secure login with valid credentials	Successful	Pass
UT-003	Skin Image Analysis	Image processed and results returned	Successful	Pass
UT-004	Personalized Recommendation	Relevant products suggested	Successful	Pass
UT-005	Add to Cart Functionality	Product added to cart correctly	Successful	Pass
UT-006	Checkout Process	Order placed with correct total	Successful	Pass
UT-007	Admin Product Management	Product added/updated/deleted	Successful	Pass

6.1.3 Functional Testing

Functional testing verifies whether **Thesara Cosmetics** meets business requirements by ensuring that each feature performs as expected.

Table 6.4 Functional Testing

Test Case ID	Test Case Name	Expected Results	Actual Result	Pass/Fail
FT-001	User Registration & Login	User can register and login securely	Successful	Pass
FT-002	Skin Image Upload & Analysis	AI analysis works and shows results	Successful	Pass
FT-003	AI Chatbot Interaction	Chatbot responds accurately	Successful	Pass
FT-004	Product Browsing & Recommendations	Personalized suggestions displayed	Successful	Pass
FT-005	Shopping Cart & Checkout	Order placed successfully	Successful	Pass
FT-006	Order Tracking	Real-time status visible	Successful	Pass

FT-007	Admin Dashboard Functions	Admin can manage products and users	Successful	Pass

6.1.4 Integration Testing

Integration testing evaluates the interactions between different modules of **Thesara Cosmetics,** ensuring seamless data flow across components.

Table 6.5 Integration Testing

Test Case ID	Test Case Name	Expected Result	Actual Result	Status
IT-001	Frontend-Backend Communication	Data displayed correctly	Successful	Pass
IT-002	Database Integration	All CRUD operations work	Successful	Pass
IT-003	AI Module Integration (OpenCV)	Skin analysis results saved correctly	Successful	Pass
IT-004	Payment Gateway Integration	Secure transaction processed	Successful	Pass
IT-005	Role-Based Access Control	Proper permissions enforced	Successful	Pass
IT-006	Responsive Design Across Devices	Layout adapts correctly on all screen sizes	Successful	Pass

Chapter 7

Conclusion and Future Work

7 Conclusion and Future Work

7.1 Conclusion

Thesara Cosmetics successfully demonstrates the integration of artificial intelligence with e-commerce to address a significant real-world problem in the skincare industry: the lack of personalized product recommendations. By combining computer vision for skin image analysis, natural language processing through an AI-powered chatbot, and a robust Laravel-based e-commerce platform, the system delivers a comprehensive, user-centric solution that goes beyond traditional online shopping.

Users can securely register, create detailed skincare profiles, upload facial images for AI-driven skin condition detection (such as acne, dryness, or pigmentation), receive personalized product recommendations, browse and filter products, manage their shopping cart, complete secure payments, track orders, and provide feedback. The admin dashboard provides efficient tools for product management, user oversight, order processing, inventory control, and analytics, ensuring smooth platform operations.

Key achievements of the project include:

- Accurate and dynamic AI-based skincare recommendations powered by OpenCV for image analysis and Open Router API for conversational guidance.
- A responsive, intuitive user interface built with Bootstrap and Laravel Blade templates that works seamlessly across devices.
- Secure authentication, role-based access control, and integration with payment gateways.
- A well-designed relational database (MySQL) with normalized tables supporting all core functionalities.
- Thorough testing (manual, unit, functional, and integration) that validated the system's reliability, usability, and performance.

The development followed an Iterative and Incremental model combined with Agile practices, allowing continuous feedback from the supervisor, co-supervisor, and stakeholders. This approach ensured that both functional and non-functional requirements outlined in the SRS were effectively implemented while maintaining modularity, scalability, and maintainability.

Thesara Cosmetics not only bridges the gap between generic e-commerce platforms and personalized skincare solutions but also showcases the practical application of AI technologies (computer vision and NLP) in a consumer-facing system. It reduces the risk of mismatched purchases, enhances customer satisfaction, and provides administrators with powerful management capabilities. Overall, the project successfully fulfills its objectives and serves as a strong demonstration of how AI can transform traditional e-commerce into an intelligent, personalized experience.

This Final Year Project has significantly contributed to our technical growth in web development (Laravel), artificial intelligence, database management, software engineering principles, and project management. It has prepared us for real-world software development challenges and highlighted the importance of user-centric design in AI-driven applications.

By combining computer vision for skin image analysis, natural language processing through an AI-powered chatbot, and a robust Laravel-based e-commerce platform, the system delivers... as supported by growing body of research in AI dermatology and personalized beauty tech (Li et al., 2022; Khan et al., 2025).

7.2 Future Work

While the current version of Thesara Cosmetics provides a solid foundation, several enhancements can further improve its capabilities and reach:

Mobile Application Development: Build a cross-platform mobile app using Flutter to improve accessibility and provide native experience for Android and iOS users.

Advanced AI Improvements:

- Integrate more sophisticated machine learning models trained on larger dermatology datasets for higher accuracy in skin analysis.
- Implement real-time skin condition tracking and progress monitoring over multiple uploads.
- Add AR-based virtual try-on features for skincare products.

Enhanced Personalization and Features:

- Subscription-based personalized skincare kits generated from ongoing AI analysis.
- Advanced recommendation engine using collaborative filtering and user behavior analytics.
- Multi-language support (full Urdu and English localization).

Additional Integrations:

- Social media login and sharing options.
- Push notifications for offers, order updates, and skincare tips.
- Integration with more payment gateways and delivery partner APIs for real-time tracking.

Security and Scalability:

- Implement advanced security measures such as two-factor authentication and enhanced data encryption.
- Optimize the system for higher scalability to support thousands of concurrent users using cloud services (AWS or Digital Ocean) with load balancing.

Business and Marketing Features:

- Analytics dashboard with predictive insights for sales and inventory.
- Loyalty program and referral system.
- Moderated community features or skincare blog integrated with AI recommendations.

Addressing these future enhancements will transform Thesara Cosmetics from a functional prototype into a production-ready, market-competitive platform capable of making a meaningful impact in the beauty and skincare e-commerce sector.

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